Credit Allocation Mix in Peer-to-Peer Lending: A Network Study on Bondora

Social Network Analysis - Group 7

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I Abstract

In this study we investigate the behavioural phenomena that influence individual borrowers' choice of loan purposes. In the first study, we evaluate how loan counts within each loan purpose choice are influenced by borrowers' individual characteristics by asking "*Do borrowers vary in their loan purpose type and frequency based on the individual and loan characteristics?"*. Using the Quadratic Assignment Procedure (QAP) linear regression framework. Our first hypothesis asserts that borrowers sharing the same credit rating should choose similar loan purposes, but with a preference for a lower loan count. We found that while this is a significant effect, the impact on loan count is minimal. The second asserts that borrowers sharing the same occupation area should share loan purposes with a lower count as well. Here, we do not find a meaningful effect. The second study examines which behavioural phenomena affect how borrowers decide which combination of loan purposes to request. We ask, *"What behavioural mechanisms affect the structure of borrower-to-loan-use connections"*. With the use of Exponential Random Graph Models (ERGM), we first determine whether there is co-affiliation between borrowers who overlap in their loan choices. We find that there is no systematic aggregate overlapping effect. Further, we evaluate whether there is an individual behavioural effect where borrowers minimise the number of loan types. The results indicate a systematic individual behavioural bias. Assessing whether younger borrowers have different loan purpose selections, we fail to find evidence for this assertion. Similarly, we fail to find support for the idea that there are gender differences in how borrowers choose loan types.

1 Introduction

With the advent of online peer-to-peer (P2P) lending platforms, the traditional methods of financial intermediation have been usurped by individual choice; with no stringent third parties involved in transactions, individuals have greater power in sourcing credit. This presents new realities for individuals disenfranchised by the traditional financial system (Milne & Parboteeah, 2017). Traditional financial literature shows that individually, investors exhibit behavioural biases (Agrawal, 2012; Ayal, Hochman, & Zakay, 2011) in how they choose their investments. However, numerous studies demonstrate that these behavioural biases are present among non-professional P2P borrowers. For example, Herzenstein, Dholakia, & Andrews (2011) and Lee & Lee (2012) suggest that users cluster around popular loans, exhibiting 'herding' behaviour. Literature focuses predominantly on funding success (Yao, Chen, Wei, Chen, & Yang, 2019), however, Ayal, Bar-Haim, & Ofir (2018) suggests that there is insufficient attention on how borrowers form their loan portfolios. As the authors indicate, this area of research is important in understanding why non-professional investors deviate from known models of rational investor behaviour. Our study aims to model and understand how P2P borrowers' behaviours influence what kinds of loans they acquire. In subsequent discussions, we consider investors as rational participants in credit markets, as described by theory, whereas borrowers refer to the users of P2P platforms.

Investigating behavioural biases, Ayal et al. (2018) find that investors' familiarity with specific assets can make them seem less risky, making recognisable assets more attractive. However, such assets lead to undiversified portfolios. Evidence with P2P platforms supports the familiarity bias in lenders, as Galak, Small, & Stephen (2010) demonstrate that lenders choose borrowers who are similar across demographic attributes such as gender, occupation, and ethnicity. Moreover, Amar, Ariely, Ayal, Cryder, & Rick (2011) explain that borrowers show an aversion to holding a *large number of debts,* preferring to first reduce debt count, even if they had other debts with higher interest rates. This irrational behaviour demonstrates that individual borrowers face behavioural biases on the *type* and *number* of debts they select. Based on these notions, we formulate our first study, asking:

Research Question 1: *Do borrowers vary in their loan purpose type and frequency based on their individual and loan characteristics?*

This question is answered using linear Quadratic Assignment Procedure (QAP) regressions, as we can represent similarities across loan purposes and attributes as matrices, weighted by loan purpose frequency, as inputs in the model.

Using data from LendingClub, Serrano-Cinca, Gutiérrez-Nieto, & López-Palacios (2015) study funding success, finding that the reported loan purpose is a significant factor explaining differences in default rates. Specifically, the authors argue that there is heterogeneity in the credit riskiness of individuals seeking specific loan purposes, indicating that credit rating can explain some variation in borrowers' choice of loan purpose. Considering the preference for lower loan counts as well, we formulate the following hypothesis:

Hypothesis 1: Borrowers with the same credit ratings choose fewer loans within their shared loan purpose

Supporting the familiarity bias, Ravina (2019) studied personal characteristics in P2P markets, finding that loans tend to be awarded to those sharing personal characteristics such as occupation.

As Ayal et al. (2018) imply, this can be generalised to suggest that lenders' loan grants reflect a wider network of similar borrower preferences, which can reflect the homogeneity in loan purpose observed by Serrano-Cinca et al. (2015). Therefore, we expect occupation to be a meaningful predictor. However, per Amar et al. (2011), we expect a preference for smaller loan counts within borrowers' shared occupations. Thus,

Hypothesis 2: Borrowers sharing the same occupation area tend to share fewer loans for their shared loan purpose.

Further, behavioural finance suggests that individuals' financial decisions are not independent, but rather, shaped by bounded rationality and psychological preferences (Kahneman & Tversky, 1979). In the P2P context, Ayal et al. (2018) shows that these preferences can influence how borrowers choose loan types and how this can overlap with other borrowers' decisions. Using the ERGM framework, we can identify how these behaviours manifest as structural regularities in a borrower-to-loan-use network. Therefore, we ask:

Research Question 2: *What behavioural mechanisms affect the structure of borrower-to-loan-use connections?*

The previous section outlined that the familiarity bias can lead to similarities across borrower portfolios in terms of their characteristics. However, in P2P settings, this can manifest itself as familiarity *clustering*, wherein borrowers who have previously been granted certain loan types are more likely to repeat their choices, or select other categories that appear familiar to different borrowers. Herzenstein et al. (2011), studying borrower decisions on Prosper.com, identified a 'strategic herding' behaviour, where individuals, **in aggregate,** gravitate towards popular or socially-validated loans. Considering that loan popularity is directly observable through its number of bids, we assert that that the social validation of a loan purpose for one borrower positively reinforces the selection of that same loan purpose for other, **co-affiliated** borrowers.

In network terms, this aggregate herding behaviour generates a structural propensity for four-cycles to form, resulting in closed paths between overlapping nodes of each of the bipartite partitions, as suggested by Koskinen & Edling (2012). This structure represents the fact that two borrowers connected to a common loan purpose have a higher likelihood of also connecting to a *second* common loan purpose. We capture this specific form of bipartite closure through the ERGM term `cycle(4)`, formulating:

Hypothesis 3: Co-affiliated borrowers are more likely to overlap on a second loan purpose.

Further, Ayal & Zakay (2009), expand on the theory of Debt Account Aversion with the Theory of Perceived Diversification, explaining that **individual** borrowers' distress increases with not only with number of debts, but also the perceived *distinctiveness* of each debt type, suggesting that in the P2P context, debt that is mentally differentiated by its purpose would also be seen as distressing. Therefore, we assert that if borrowers do not diversify, then they try to minimise loan distinctiveness by requesting loans of a single type. This can be captured by the `b1degree(1)` term, which models whether borrowers are only linked to one loan purpose. Therefore, we formulate:

Hypothesis 4: Borrowers exhibit preferences for minimal debt distinctiveness, selecting one loan purpose.

Notwithstanding, credit choices relate to non-structural factors. Cooper, Gorbachev, & Luengo-Prado (2023) show that younger borrowers face constraints in the types of loans they can access in typical credit markets. The authors argue that this constraint shapes how these borrowers form loan portfolios, often concentrating around different use cases such as education, personal consumption, or small business loans. Considering the systemic differences in how age affects credit consumption, we assert that younger borrowers have different loan purpose portfolios. The ERGM term `b1cov("age")` can capture this by assessing how changes in age affect how borrowers form connections with different loan types. Therefore:

Hypothesis 5: Younger borrowers have different loan purpose mixes than older ones.

Alongside age, the borrower's gender is known to be a significant factor in influencing credit use decisions. Aliano, Alnabulsi, Cestari, & Ragni (2023) studied the role of gender and other factors on borrowers' probability of default on Bondora.com, finding a persistent gender effect on different loan purposes; women tend to default less on health, home, and business loans. Croson & Gneezy (2009) show that these differences can be due to risk perception and self-selection effects, where women exhibit greater risk aversion, leading to different borrowing patterns. We expect that these differences in risk tolerance and perceived creditworthiness by lenders affect loan purpose composition in terms of gender. The term `b1factor("gender")` captures this effect by assessing whether borrowers' connections with loan purposes differ across genders. We would find support for this idea given a significant, non-zero effect estimate. Therefore:

Hypothesis 6: Gender differences lead to different loan purpose mixes.

Table 1: Hypotheses related to Study 2

Hypothesis	Term	Explanation
H_3^a : Co-affiliated borrowers are more likely to overlap on a second loan purpose	*cycle(4)*	This captures the tendency of borrowers who are already affiliated through their connections to the same loan type to have connections to another type. With this term we capture **aggregate** borrower behaviour.
H_4^a : Borrowers exhibit preferences for minimal debt distinctiveness, selecting one loan purpose	*b1degree(1)*	This captures the tendency of borrowers to prefer minimal distinctiveness in their loan purposes, preferring to hold only one distinct type to minimise their perceived risk. With this term, we capture **individual** borrower behaviour rather than an aggregation phenomenon.
H_5^a : Younger borrowers have different loan purpose mixes than older borrowers	*b1cov("age")*	This captures the tendency for younger borrowers to choose different kinds of loans based on how differently they consume credit.
H_6^a : Gender differences lead to different loan purpose mixes	*b1factor("gender")*	There are systemic differences in what kinds of loan purposes are preferred the genders due to their perceived riskiness and creditworthiness.

This study makes a specific contribution to behavioral finance by leveraging Linear QAP Regression and the ERGM to empirically isolate the structural and individual behavioral biases, such as familiarity clustering and debt account aversion, that drive non-professional P2P borrower portfolio construction, thereby offering actionable insights for platform risk management and regulatory policy stakeholders.

Following this section, we outline the methodology, explaining our dataset and network construction. To facilitate this, descriptive statistics are analysed for both the dataset and resulting network. Subsequently, we elaborate on our choice of models, discussing how network models can be used to conduct the study. Then, modelling results are shown and explained to evaluate our hypotheses. Simultaneously, the robustness of the study is evaluated through the models' diagnostics. Finally, we arrive at our conclusions, summarising the study, its results, and implications. Supporting items such as the source code and specific data processing steps are shown in section A.

2 Dataset

The study utilises publicly-available data from Bondora, a European P2P lending platform, primarily operating in the Baltics and Spain. The dataset contains detailed information on defaulted and non-defaulted loans granted to users between February 2009 and July 2021. Specifically, contained is a range of numeric, binary, categorical, and time-series attributes across `85,087` unique users and `179,235` individual loans. Since the company's API is no longer accessible, we utilised a publicly available repository compiled by Manu Siddhartha (2021) on kaggle.com. The data compiler made a series of API calls to collect the data. Following this, the dataset was processed according to Appendix Section A.1.

2.1 Pre-Processing & Network Formation

Initially, the dataset was filtered to loans between 2014 and 2016 since the data is incomplete during other periods. Since we are studying interactions across two distinct set of nodes, a bipartite network was formed with unique users as the first partition and the purpose of loans as the second. This allows us to evaluate how borrowers interact with loan purposes; a one-mode projection onto users is not meaningful for P2P networks as it produces no discernible structure. For model efficiency and performance, we randomly sampled `500` individuals from the larger sample. Ultimately, we have `500` nodes of users and `9` nodes of loan purposes, in the first and second partitions, respectively.

2.2 Descriptive Statistics & Preliminary Analysis

Table 2 summarises the numeric variables for the reduced and larger samples. First, borrowers take out relatively small loans with fairly high interest rates, possibly reflecting that the average borrower is deemed fairly risky. The average loan is quite long, at approximately `46` months out of a maximum of `60`. Figure 18 shows that the largest credit category is `HR`[1], which is the lowest possible rating. So, the typical borrower is rated as relatively uncreditworthy. Moreover, the largest

[1]Bondora uses their own proprietary credit rating system where the best rating is AA and worst is HR (Bondora, 2024). The company bases these ratings on their calculated expected probability of loss on the loan. For example, someone rated AA ranges from a expected loss of 0% to 2%, whereas for HR this is 25% - >25%.

categories of loan purpose are `Other` and `Home Improvement`, suggesting these are largely personal. Additionally, Figure 20 shows that the biggest borrower group is from Estonia. However, Spain and Finland are also highly represented. Conversely, Slovakia has minimal representation. We observe minimal differences between the random sampling and the larger sample of individuals, per Table 2 and Figure 18. However, the study may be biased by the fact that the dataset contains only granted loans, rather than all bids for loans. This means our inferences about how borrowers behave is limited to successful bids. Further, the time period is relatively distant, meaning that recent structural changes to the P2P sector cannot be accounted for.

Table 2: Descriptive Statistics of Numeric Variables

DataFrame	Variable	Mean	Median	Std. Dev.	Min	Max
Initial Sample	Amount	2652.01	2125	2151.28	115	10630
Initial Sample	Interest	35.94	31	24.77	7.62	263.63
Initial Sample	Age	38.53	37	11.4	19	70
Initial Sample	LoanDuration	45.5	60	17.53	3	60
Reduced Sample	Amount	2637.64	2112.5	2112.2	170	10630
Reduced Sample	Interest	35.85	30	27.24	10.17	253.08
Reduced Sample	Age	38.73	37	11.34	21	69
Reduced Sample	LoanDuration	46.23	60	17.35	3	60

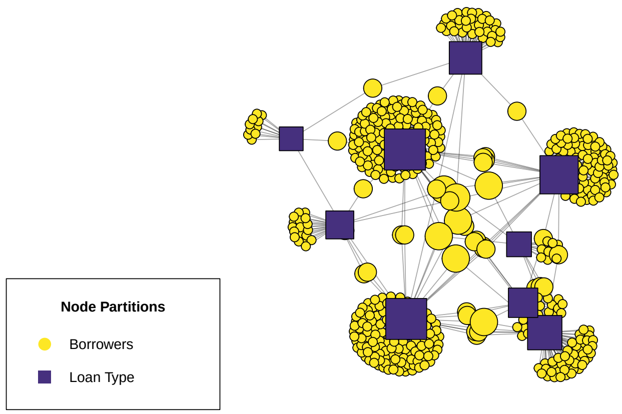
Table 3 shows a network density of `0.12`, indicating that only 12% of all possible borrower–loan-type connections exist. This density suggests that borrowers typically engage with few loan types, reflecting specialization in borrowing behaviour. The centralization value of `0.47` further indicates an uneven distribution across loan types, where a few popular categories attract many borrowers while most remain peripheral. The mean distance of approximately three implies that borrowers are closely connected, typically separated by only three-to-four loan-use steps.

Centrality distributions reinforce this structure as shown in Figure 2. Betweenness is highest among loan-type nodes, though several borrowers also bridge distinct loan clusters. Closeness centrality remains low but scattered, suggesting localised clustering constrained by the bipartite structure. Degree distributions show wide variation among loan types, consistent with a hub-like structure dominated by a few popular uses. However, the second partition has relatively small degree dispersion. Finally, the homogeneous eccentricity distribution indicates that no borrowers are highly isolated, reflecting a compact and moderately cohesive network overall.

Table 3: Basic Network Descriptive Statistics

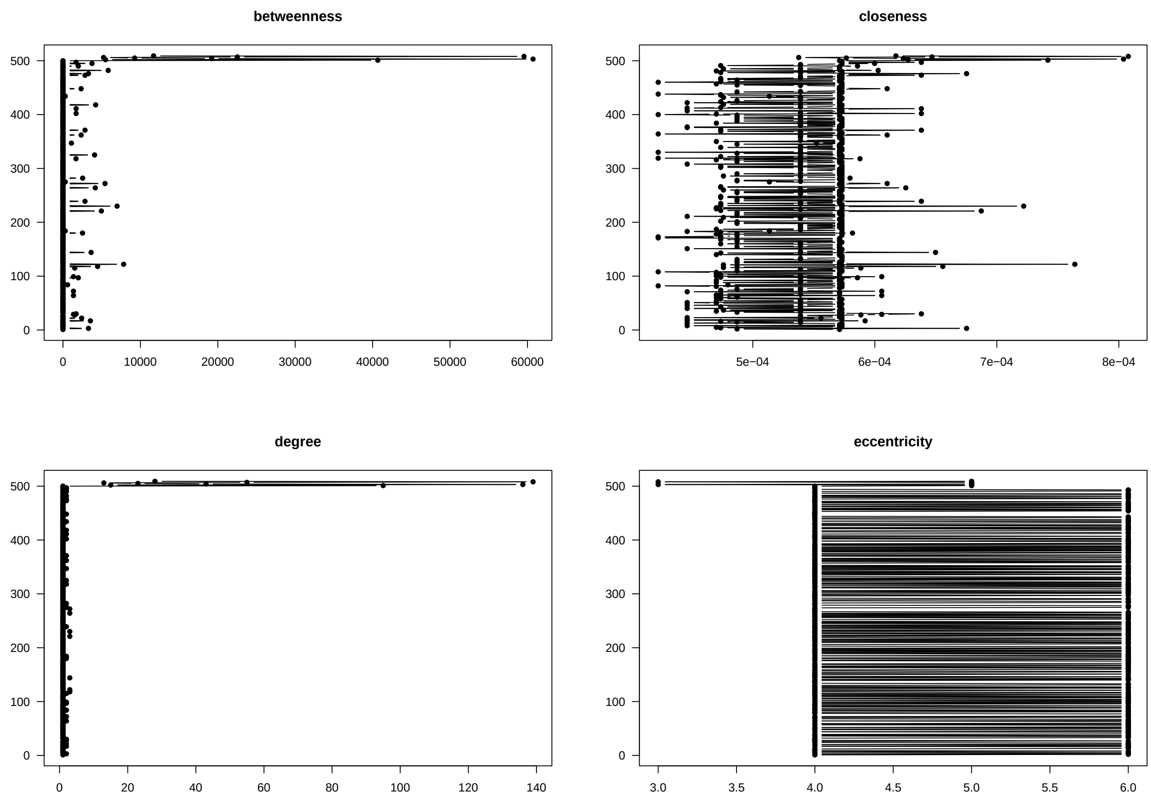
Statistic	Measure
Vertex Count	509.00
Edge Count	547.00
Density	0.12
Centralization	0.47
Mean Distance	3.66

Figure 1: Network Visualisation



Note: Each colour represents a different partition of the network. The partitions' nodes are scaled to their relative degrees. Users with many different loan types are larger than those with single loan purposes. Similarly, the size of second partition nodes represents how many users belong to each.

Figure 2: Network Centrality Distribution Plots



3 Research Rationale

We investigate borrower behaviour, however, the studies are differentiated in their scope; the first study observes dyad-level relations among borrowers to understand how shared attributes affect borrowers' behaviour. This is inherently useful because behavioural finance aims to understand how individuals behave in *aggregate* (Ayal et al., 2018). Typically, the Quadratic Assignment Procedure is selected for dyad-level observations because basic linear regressions such as Ordinary Least Squares are fundamentally flawed and biased when using non-independent observations (Simpson, 2001).

Recognising this, studies such as Zuo (2015) assume that groups of borrowers have dependencies, and accordingly, use clustering methods such as K-Means or Fuzzy Clustering to understand behaviour. However, these methods are unsuitable when we require a distribution of estimates to make inferences from our results. Moreover, studying individuals as dyads rather than large clusters provides finer granularity in results. Therefore, QAP regression is the most appropriate choice for this study: it allows us to control for the non-independence among dyads, derive a statistical distribution of estimates through matrix permutation, and draw valid inferences about the relationship between borrower similarities and loan-use behaviour. This makes it both theoretically and methodologically well-suited to our research objectives and dataset.

The second study extends the analysis from dyadic similarities to the structural formation of borrower-to-loan-use connections, which can identify the behavioural mechanisms that influence borrowers' specific loan purpose mixes. While QAP can account for exogenous influences, it cannot analyse higher-order network patterns such as clustering while accounting for simultaneous interactions between borrowers and loan purposes. Using ERGM allows us to analyse how local behavioural tendencies aggregate to a global network structure.

Recent evidence highlights the need to focus on structural approaches in P2P lending. Liu, Baals, Osterrieder, & Hadji-Misheva (2024) studied borrowers' centrality with the Bondora P2P dataset, modelling edges through borrower-to-borrower similarity metrics. Ultimately, they find that a borrower's position is highly significant for their probability of default, using centrality as an exogenous variable in a logistic regression. In contrast, ERGM allows us to move away from predictive association and instead account for endogenous factors, allowing us to understand the likelihood that borrowers' selections of loan purposes is random. Consequently, we can better understand how behavioural phenomena such as familiarity and debt account aversion can translate into *systemic* patterns of credit use.

As we aim to understand how two distinct groups of nodes interact, a bipartite approach is highly suitable. Specifically, Koskinen & Edling (2012) shows that in bipartite networks, terms such as `cycle(4)` can be effective in capturing the idea of grouping across partitions, as in hypothesis 3, whereas it would fail in a fully-connected one-mode projected networks. Furthermore, the flexibility of ERGM also allows for modelling exogenous impacts such gender-based differences in loan purpose mixes, as in hypothesis 6, and covariate age effects as in hypothesis 5.

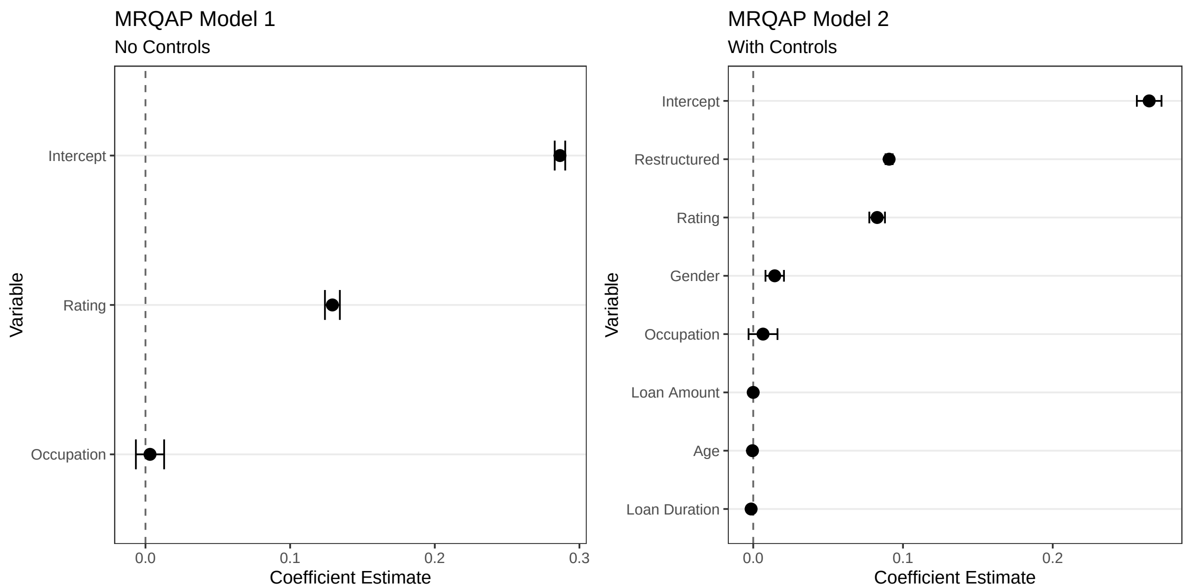
4 Results

4.1 Study 1 - QAP Linear Regression

The first study is assessed with two QAP linear regression models, as shown in Figure 3 and Table 4. The first model includes only the main predictors whereas the second controls for borrowers' age, gender, loan amount, loan duration, and whether the loan was restructured. Differences across these models can indicate the strength of omitted variable bias.

Assessing hypothesis 1, in the unrestricted model, borrowers sharing the same credit rating once leads to a `0.083` increase in the number of loans they share within the same loan purpose category, holding all other effects fixed. While this effect is statistically significant at 1%, notably, it is not economically significant because the effect magnitude is very small. In essence, having the same credit rating as another borrower does not mean that the borrower will have meaningfully different amounts of loans taken out for the same reason. Without controls, this effect was slightly larger, meaning that other predictors capture some of this effect. Our results corroborate aspects of Serrano-Cinca et al. (2015); similarly, we find that there is a non-zero effect of creditworthiness on loan purpose, however, they find a stronger effect. This can be due to borrowers' aversion to holding large numbers of debt, as explained by Ayal et al. (2011), where borrowers attempt to minimize debt counts independently of the loan purpose type.

Figure 3: QAP Coefficient Plots



Further, assessing hypothesis 2, borrowers sharing the same occupation area does not have a significant impact on the outcome. This indicates that the number of loans taken out for the same reason as another borrower does not vary, given that pairs of borrowers have the same kind of job. The omission of control variables does not change this effect, meaning that its insignificance is not due to other correlated effects. The result is contrary to Ravina (2019), who found that certain occupation areas have a higher likelihood of loan success. As the author states, these effects are likely present in markets with sophisticated decision makers. It is possible that Bondora lenders do not conform to this standard. Alternatively, the familiarity bias lenders exhibit on shared personal characteristics may not generalise to similarities across borrower characteristics, as we have argued.

Observing the controls, we see that loan amounts and whether the loan were restructured are statistically significant at 5%, but not economically meaningful. Moreover, the model diagnostics show an $R^2 = 0.035$, suggesting that the linear QAP only explains 3.5% of the variation observed in the data. This is rather small, indicating that the linear model is not a good fit for this specific study. We argue that borrowers are not biased against large debt counts per loan purpose type as *dyads*, but rather, as part of a more complex network structure. This is indicated by the low R^2, showing that this dyadic relationship cannot be captured as a linear effect.

4.2 Study 2 - ERGM

The second study is assessed with four iterative models shown in Figure 4 and Table 5; the fourth is considered to be the full, unrestricted model. In each iteration, additional terms are added and diagnosed in Appendix Section A.4 by the Goodness of Fit (GOF) and Markov chain Monte Carlo (MCMC) simulations. We first interpret model estimates, their relation to our hypotheses, and conclude by evaluating the validity of the fitted models.

In Model 4, the inclusion of `b1factor("gender")` has pushed the `edges` term towards zero, making

Table 4: QAP Modelling Results

	M1 | No Controls	M2 | Controls
Intercept	0.287^{***}	0.264^{***}
	(0.002)	(0.004)
Rating	0.129^{***}	0.083^{***}
	(0.003)	(0.003)
Occupation	0.003	0.007
	(0.005)	(0.005)
Loan Amount		-0.000^{**}
		(0.000)
Age		-0.001
		(0.000)
Gender		0.014
		(0.003)
Loan Duration		-0.001^{**}
		(0.000)
Restructured		0.091^{***}
		(0.001)
R-squared	0.010	0.035

$^{***}p < 0.001$; $^{**}p < 0.01$; $^{*}p < 0.05$

it insignificant. This suggests that the combination of individual borrowers' loan purpose formation is likely random in the presence of other terms, indicating that the edge formation effect is instead captured jointly by the rest of the terms. We consider this a good result since it suggests that our final model may be fully saturated.

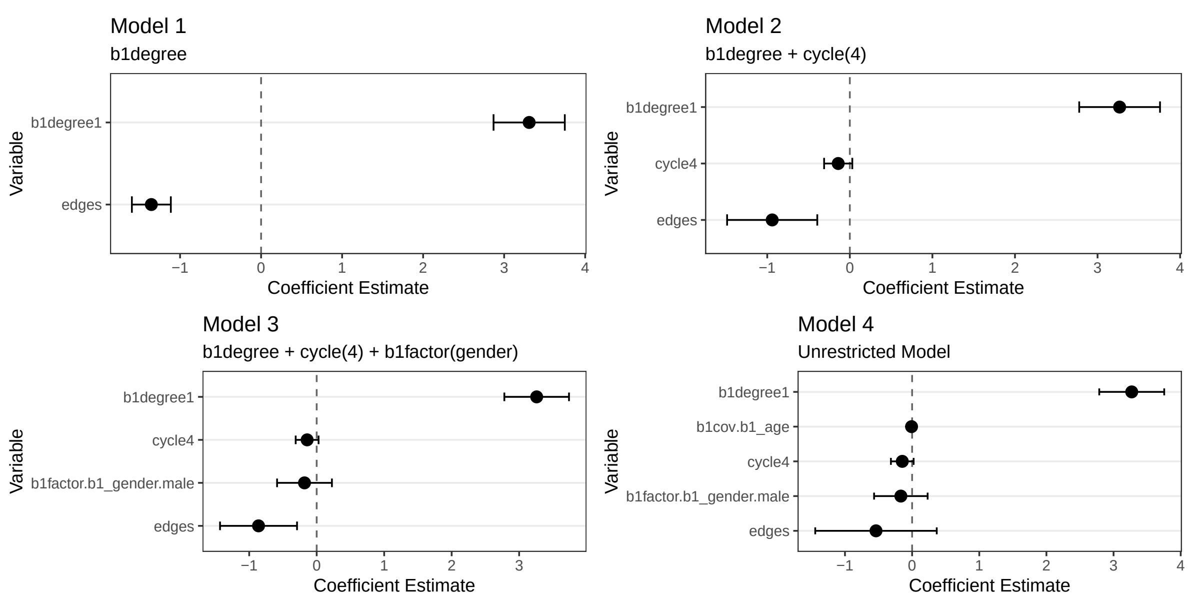
Evaluating hypothesis 4, we determine whether borrowers exhibit a preference for minimal debt distinctiveness by choosing only one loan type. In Model 4, `b1degree(1)` is highly significant and positive with a log-odds of `3.271`, suggesting that the odds of a borrower connected to exactly one loan purpose is approximately `26.34` times higher than the odds of observing a configuration where the degree is not one. In other words, in this Bondora sample, it is highly likely that borrowers select only one loan type. This effect size or error has not deviated strongly across the four model permutations, indicating its stability. This result shows support for the Theory of Perceived Diversification by Ayal & Zakay (2009) because on an **individual level,** borrowers prefer to hold minimal *distinct* debt types.

Further, the third hypothesis asserts that borrowers affiliated by a common loan purpose overlap on a second loan purpose, reflecting familiarity bias on an **aggregated** level. In Model 4, four-cycles are insignificant, meaning that we have not found sufficient evidence to reject the null hypothesis that there are no systematic co-affiliation effects. In other words, borrowers who are affiliated by the same loan purpose do not overlap on a second loan purpose. This finding is contrary to Herzenstein et al. (2011), who argues that borrowers aggregate around popular, or familiar loans. This could be explained by the fact that the authors examine auction bids, rather than winning bids, as we have. The behaviours or characteristics of bidders, compared to winners, may be fundamentally different.

Evaluating attribute-level effects, we see that neither age nor gender has a significant impact on loan purpose selection by borrowers. Specifically, we do not find sufficient evidence to conclude that borrowers being younger has any influence over how they select their loan purposes. This indicates a lack of support for hypothesis 5, which asserts that younger borrowers should have different loan purpose combinations than older borrowers. In this vein, we cannot corroborate Cooper et al. (2023), however, this is likely due to the fact that their study focuses on credit consumption of U.S. young adults, which is fundamentally different than credit consumption in Europe Jentzsch & Riestra (2006). Moreover, they studied traditional credit markets such as personal credit lines from banks rather than peer-to-peer lending services, which also vary fundamentally.

In the same vein, we do not observe sufficient evidence to determine that borrowers' gender affects how they select their specific loan purposes. Similarly, this is contrary to hypothesis 6, which states that there is a gender difference in how borrowers select loan purposes. The lack of effect is contrary to Aliano et al. (2023), who finds gender to be a significant predictor of loan purpose heterogeneity. They argue that gender differences across credit risk perception and self-selection effects are responsible, however, it is likely that these behavioural factors are already captured by the `b1degree(1)` term; as Ayal & Zakay (2009) argued, similar risk-related phenomena are responsible for Prospect Theory, which in turn drives the Theory of Perceived Diversification.

Figure 4: ERGM Coefficient Plots



Evaluating model fit, we must first evaluate the MCMC simulations seen in Figure below. Observing Model 4, we see that all terms have mixed sufficiently well. While `edges`, `b1degree(1)`, `b1factor("gender")`, and `b1cov("age")` are well centred at zero, `cycle(4)` is slightly right-skewed, indicating minor mixing issues. Additionally, the trace plots appear to resemble a "fuzzy" shape, supporting this. We find this diagnostic acceptable.

Further, the GOF measures indicate how well simulation iterations match our observed network.

Table 5: ERGM Modelling Results

	Base ERGM	b1degree(1)	cycle(4)	Age	Gender
edges	-1.978^{***}	-1.354^{***}	-0.940^{***}	-0.862^{**}	-0.538
	(0.046)	(0.123)	(0.279)	(0.291)	(0.462)
b1degree1		3.309^{***}	3.267^{***}	3.260^{***}	3.271^{***}
		(0.224)	(0.250)	(0.244)	(0.247)
cycle4			-0.141	-0.141	-0.147
			(0.087)	(0.086)	(0.086)
b1factor.b1_gender.male				-0.181	-0.168
				(0.207)	(0.203)
b1cov.b1_age					-0.008
					(0.009)
AIC	3325.828	2674.435	2676.439	2672.013	2674.852
BIC	3332.238	2687.255	2695.668	2697.652	2706.901
Log Likelihood	-1661.914	-1335.218	-1335.219	-1332.006	-1332.426

$^{***}p < 0.001$; $^{**}p < 0.01$; $^{*}p < 0.05$

Across the GOF measures, p-values close to `1` resemble a close match with reality. All geodesic distances except for `Inf` and `2` are well-modelled. The former is expected, as our bipartite configuration restricts any possibility of links within each partition, meaning that `Inf` distances can never be modelled. For `b1degree(1)`, all degrees except for zero and two are highly insignificant. It is expected that zero degrees is poorly modelled because our observed network has no isolates. Degree three is arguably poorly modelled since there are only `7` observations of it, limiting the simulations. The term `b2degree()` is not modelled well, however, this term is very difficult to capture statistically; there is no meaningful statistical variation with respect how many borrowers each loan purpose has, so the GOF test cannot reliably evaluate it. Moreover, edge-wise shared partners are all well modelled. Overall, we consider the GOF tests to be sufficient.

Finally, we evaluate the optimal statistical model configuration by observing the minimum AIC/BIC values across the models. AIC prefers Model 3 whereas BIC chooses Model 1; this is expected, as BIC penalizes less informative model terms more strongly.

```
Registered S3 methods overwritten by 'snafun':
  method        from
  plot.network  network
  print.network network
```

Figure 5: Model 4 Diagnostics Plots

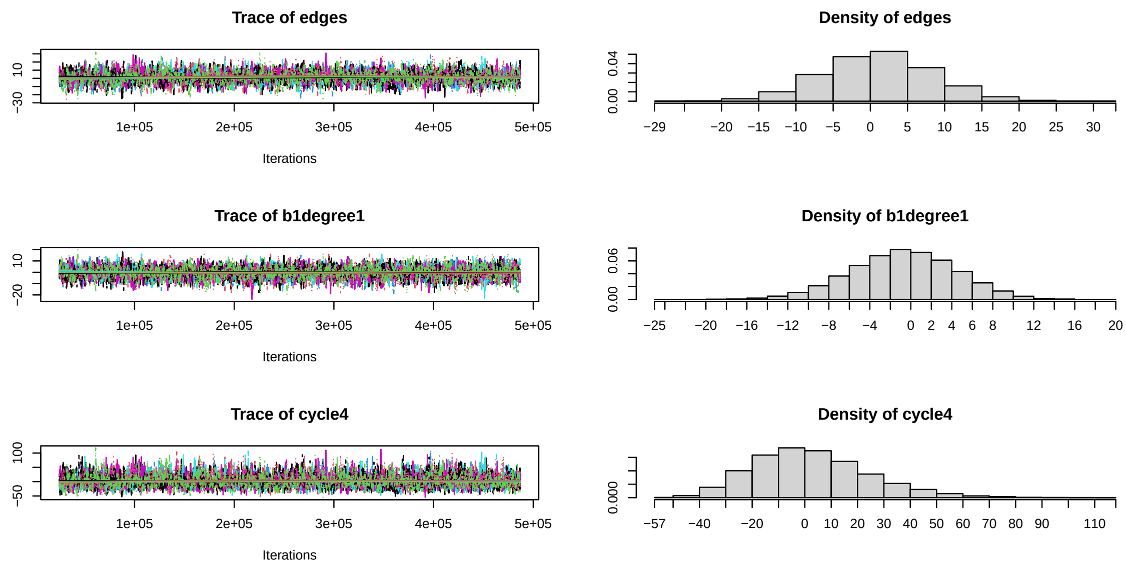


Figure 6: Model 4 Diagnostics Plots

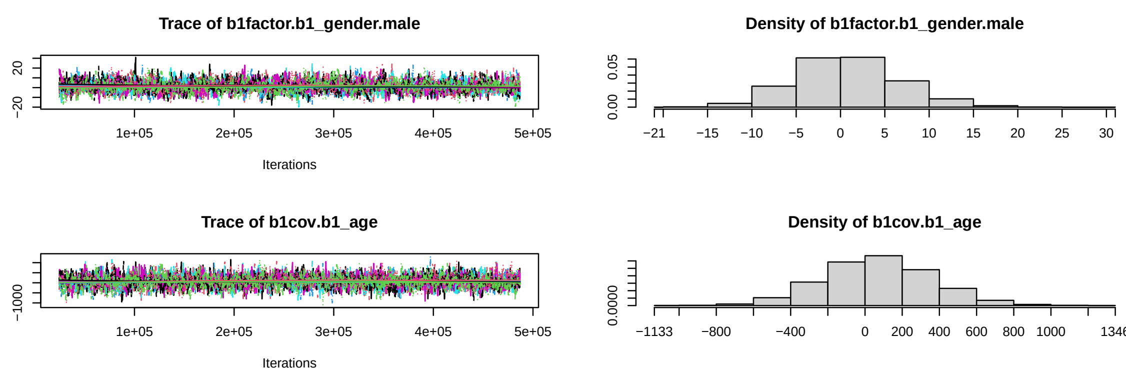


Figure 7: Model 4 Diagnostics Plots

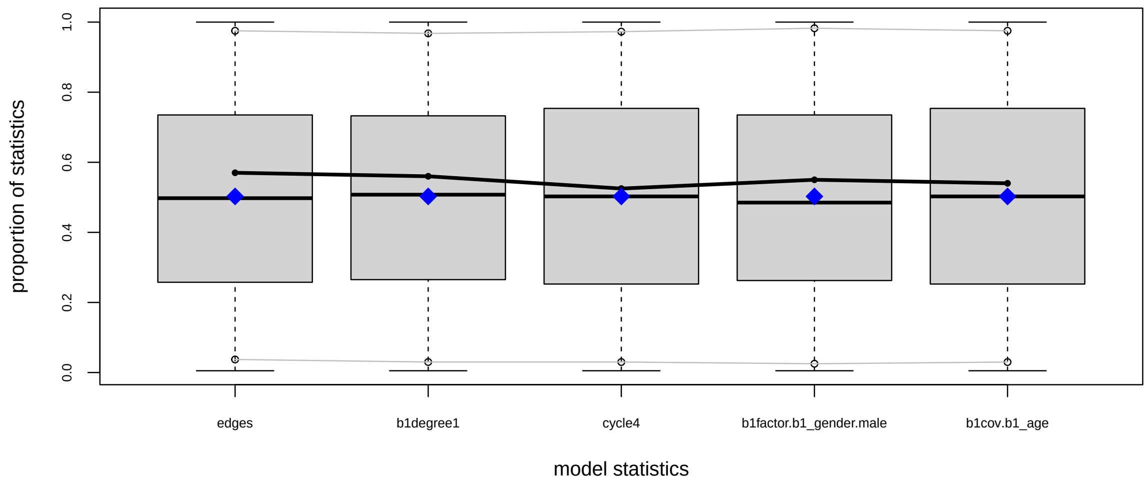


Figure 8: Model 4 Diagnostics Plots

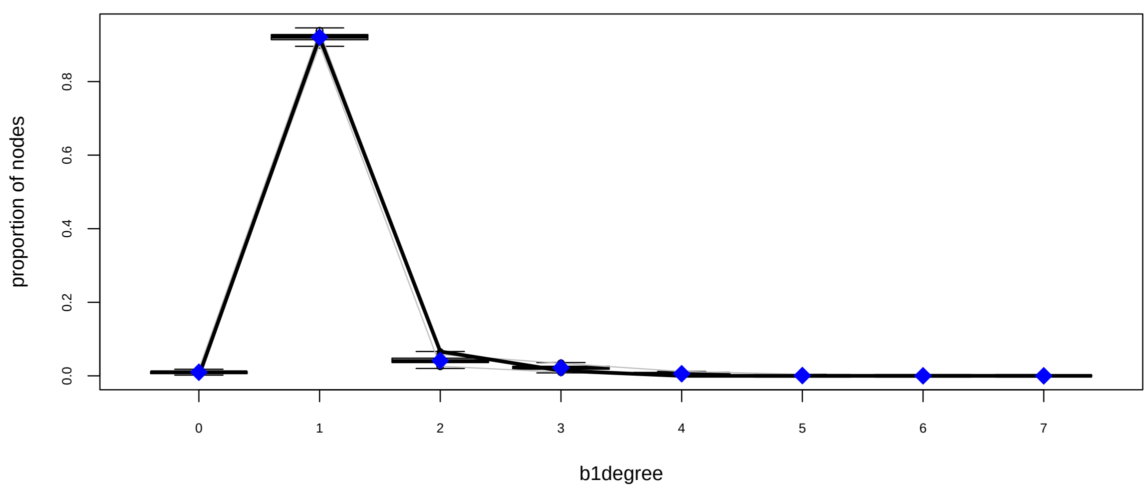


Figure 9: Model 4 Diagnostics Plots

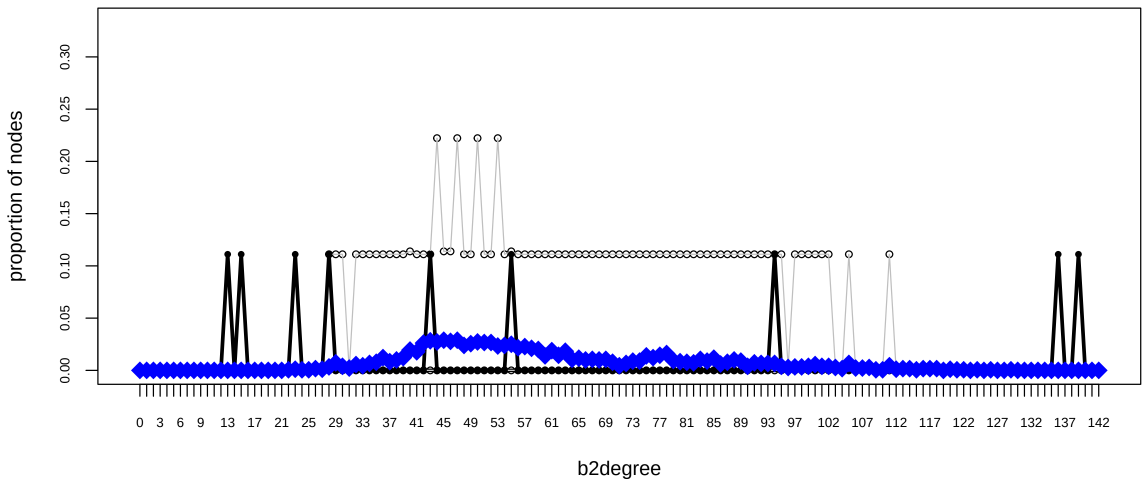


Figure 10: Model 4 Diagnostics Plots

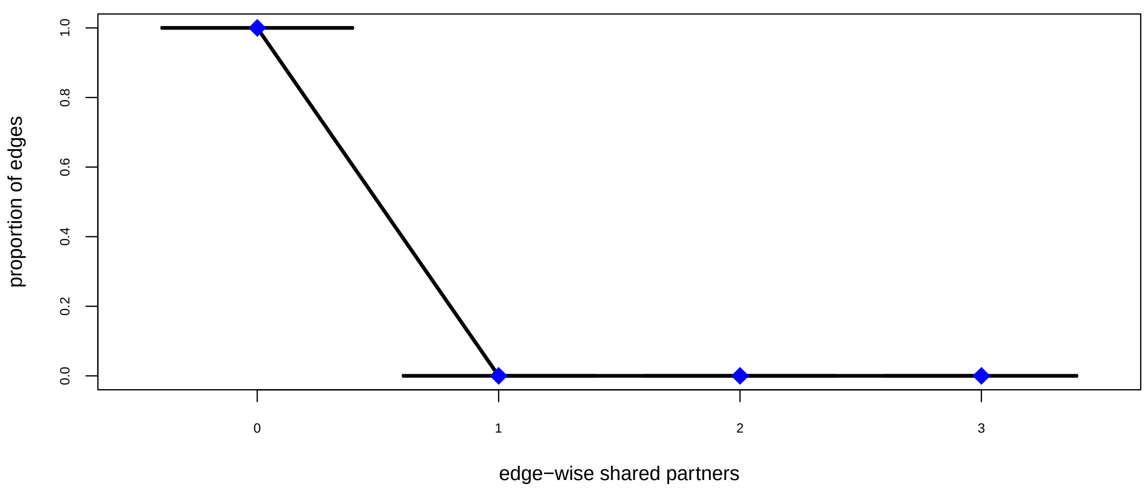
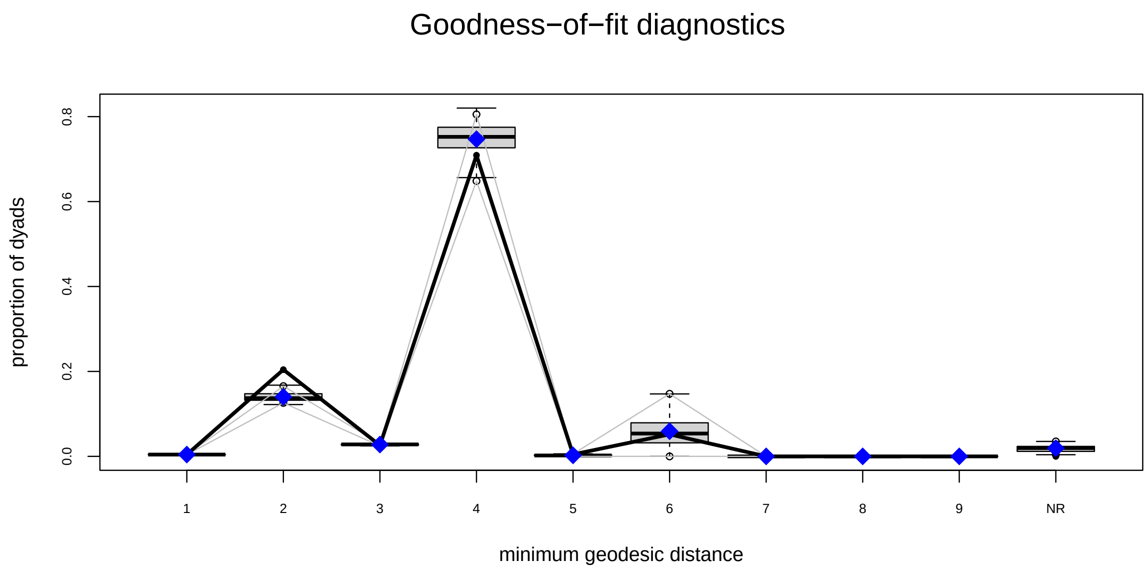


Figure 11: Model 4 Diagnostics Plots



5 Conclusion

This study aimed to assess the prevalence of behavioural biases in a peer-to-peer lending context by assessing how connections across borrowers and loan types varied; by answering the question *"Do borrowers vary in their loan purpose type and frequency based on their individual and loan characteristics"*, we can determine the extent that borrowers have an aversion to large loan counts within loan purpose categories, per Ayal et al. (2011), and by answering *"What behavioural mechanisms affect the structure of borrower-to-loan-use connections?"* in a network setting, we can see how loan purpose choices vary with structural factors.

To evaluate how loan counts are affected within loan purpose selections, we utilised the QAP linear regression framework. Within this study, borrowers are studied as dyads rather than a whole network. Firstly, it was found that borrowers having the same credit rating was significant in explaining variations in the number of loans they took out for a shared purpose, however, the impact was small in size. Further, we found that a shared occupation area between borrowers does not have a meaningful impact on loan counts. Assessing the models, we find a poor linear fit, which could indicate that dyads are not an optimal means for studying these effects. These findings do not necessarily corroborate Ayal et al. (2011) because we do not observe strong negative effects on loan counts, however, the regression diagnostics make this conclusion quite unreliable.

To evaluate specific borrower behaviours independent of loan counts, a network statistical model such as ERGM is more suitable. In this study, we differentiate between **aggregate** and **individual** borrower level effects. The results indicate very strongly that borrowers prefer to hold one loan type. This yields significant support to the Theory of Perceived Diversification by Ayal & Zakay (2009), capturing the idea that on an **individual** level, borrowers try to have the least amount of distinctiveness possible. Specifically, by holding only one loan type, they can place them in the same mental 'account', which prevents distress (Ayal & Zakay, 2009). However, we do not find that, in **aggregate,** borrowers have co-affiliation by being related to two distinct loan types. In other words, borrowers do not overlap with their choices of pairs of loan types. These two findings highlight that borrowers are more prone to individual behavioural biases rather than a 'herding' effect as described by Herzenstein et al. (2011). However, as we do not utilise auction data, it is possible that aggregate effect does not extend to how *winners* of auctions behave.

Further, examining borrowers' attribute effects, we found that neither gender nor age have significant effects on how borrowers choose loan types. Specifically, gender differences do not account for a difference in loan purpose formation. We argue that this could be due to the idea that the same gender differences that affect risk perception are also responsible for the underlying behavioural phenomena behind the Theory of Perceived Diversification. Consequently, the use of `b1degree(1)` had already captured this process. Finally, we do not find that younger borrowers have a different loan purpose mix than older borrowers. This theory was derived from Cooper et al. (2023), who studied young adults in the U.S. However, considering the fundamental differences across credit markets between the U.S. and Europe, it is likely that these behavioural effects may not apply to European borrowers across our sample of three countries. Evaluating the ERGM models, we find them to be well constructed and our diagnostics show that they largely conform to our expectations. Therefore, we consider these findings to be robust to modelling or specification error.

Arguably, the main beneficiaries of our findings are credit regulatory agencies in Finland, Estonia, and Spain, to which the majority of our borrowers fall under. Some studies indicate shortcomings in how these specific credit markets are regulated, owing their young age and online nature (Lenz, 2016). By understanding the behavioural phenomena driving borrowers' behaviour, these institutions can construct policies to minimise the risks of users utilising credit inappropriately, for example, by systematically maximizing credit counts. Further, while we have studied how credit counts and borrower selections are influenced, we have been unable to combine these phenomena together by investigating loan counts within the ERGM framework. This extension would be valuable because it would account for the shortcomings of the QAP regressions' linearity whilst also allowing for more complex behavioural phenomena to be modelled. In general, however, we have observed that researchers focus on singular P2P platforms. We recommend that studies focus on concurrent analyses of a combination of P2P platforms because individual platforms may have idiosyncrasies that make analyses difficult to generalise to the wider P2P lending market.

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A Appendix

A.1 Data Preprocessing Steps

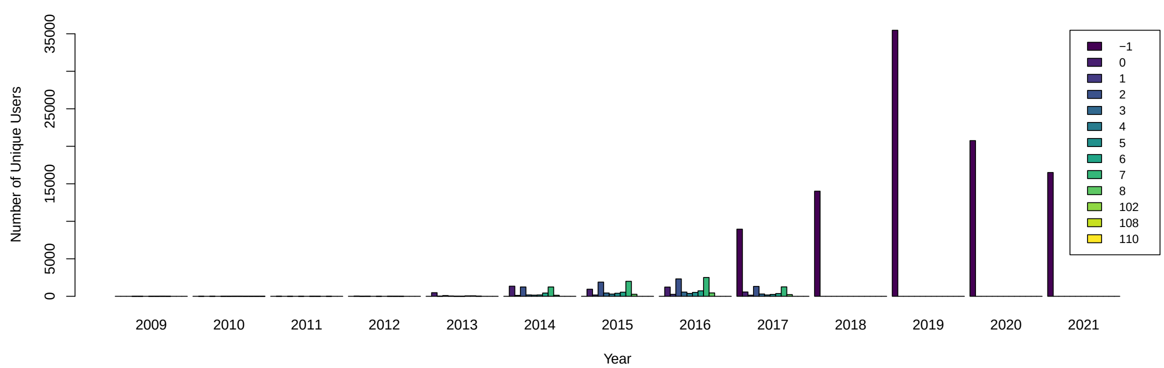
This section outlines the specific steps taken to process the data into the network object that was used in later analysis, outlining the rationale, where needed.

- Following the data import, only the following attributes were kept to make our processing steps more focused

```
keep_cols <- c("LoanId", "UserName","Age", "Gender",
              "Country", "Amount", "Interest","LoanDuration",
             "UseOfLoan", "Rating", "Restructured", "MonthlyPayment",
             "OccupationArea", "BiddingStartedOn")
```

- We removed rows with any `NA` values for the selected attributes to ensure that we have a complete dataset
- Observing Figure 12, we see that between 2017-2021, there exists many `-1` values for `UseofLoan`. This is because Bondora no longer provided this detail in later datasets. However, between 2014 and 2016 this data is available. Because of this, we restrict the time period to have the least noisy data. `BiddingStartedOn` was chosen to record the time, because the auction start date can be a good indication of activity around the loan.

Figure 12: Number of Unique Users per Loan Type per Year



- Then, we sampled `500` individuals out of the filtered dataset to ensure that our models run sufficiently well and converge. By choosing `500` individuals, we preserve importance structures within the network while adding sufficient statistical power.
- When adding labels to each loan purpose and occupation area, we keep any possible missing values because these do not indicate lower quality data. Instead, they can carry information about lenders distribute loans according to the quality of information presented by borrowers (Yao et al., 2019). Additionally, keeping missing labels is important to ensure that the edges are not superficially limited and that nodes can be isolates if they are indeed that way in reality.

- To create the network object for ERGM models:
  - We first create an incidence matrix of size $n \times m$, borrower usernames by the loan purpose, where the values are the number of loans the user obtained for each loan purpose category. These values are then turned into binary because our modelling does not support the use of weighted edges.
  - A bipartite network is created from this incidence matrix, where the first partition are unique users, and the second partition is the loan purpose associated among all loans of the user. Users can share more than more loan purpose type.
  - We add age and gender attributes to each node of the first partition
- To create the relevant networks/matrices for QAP Regressions:
  - We first re-create the loan purpose incidence matrix, however, we keep the counts as the linear regression supports this.
    * We then convert the loan purpose incidence matrix into an adjacency matrix of shape $n \times n$ through the transformation $\mathbf{X} \cdot \mathbf{X^T}$, which results in a matrix of shared loan purposes weighted by the frequency of each user's loan count within each loan purpose category.
    * We then set the diagonal values of $\mathbf{X}$ to be zero to ensure that there are no loops within the network.
  - In accordance with this procedure, we create adjacency matrices for `Credit Rating` and `Occupation Area` which will be used as the main predictors in the QAP models.
    * `Credit Rating` is a weighted adjacency matrix based on the loan count belonging to each user for each credit rating category
    * `Occupation Area` is a binary adjacency matrix based on which occupation each user has reported to have held in the past when applying for each of their loans.
  - Additionally, we create a set of control variables to improve the validity and performance of the linear regression using `Loan Amounts`, `Age`, `Gender`, `Loan Duration`, and whether the loans have been restructured.
    * `Loan Amounts` is a weighted adjacency matrix based on five bins across the range of possible loan amounts
    * `Age` is an adjacency matrix based on differences in users' ages
    * `Gender` is a binary adjacency matrix based on users' shared gender.
    * `Loan Duration` is an adjacency matrix based on the differences in users' loans average durations
    * `Restructured` is a binary adjacency matrix based on whether the users share the fact that they have defaulted on any loans in the past. The users can either have shared both defaults and non-defaults, or both.

A.2 Distributions of Variables across Samples

Figure 13: Distribution of Loan Amount across Samples

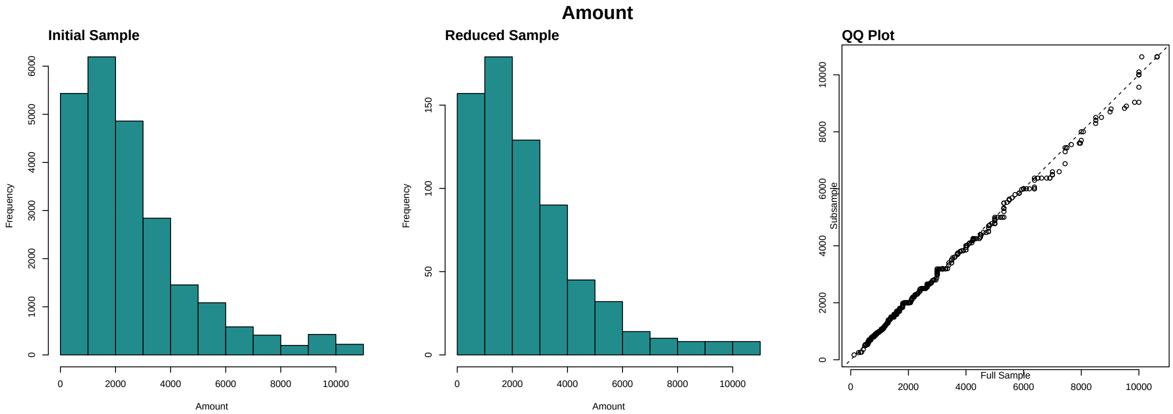


Figure 14: Distribution of Interest Rates across Samples

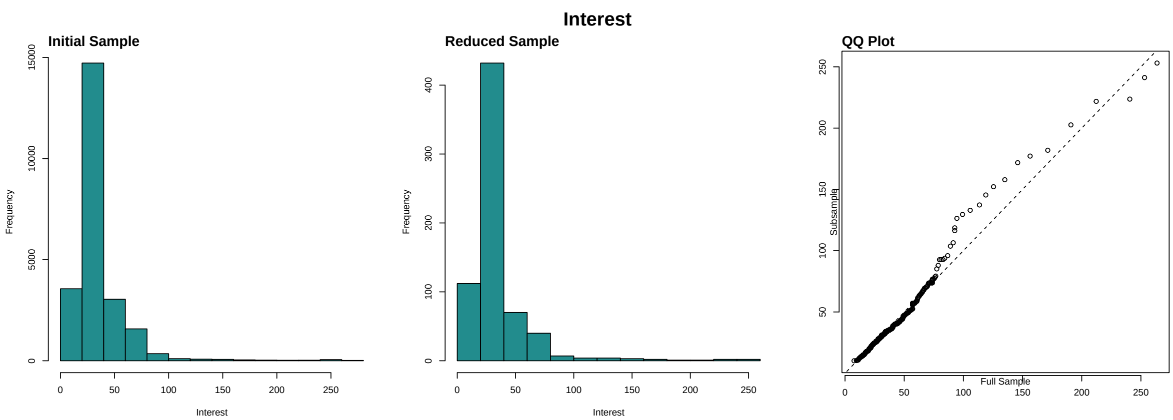


Figure 15: Distribution of Age across Samples

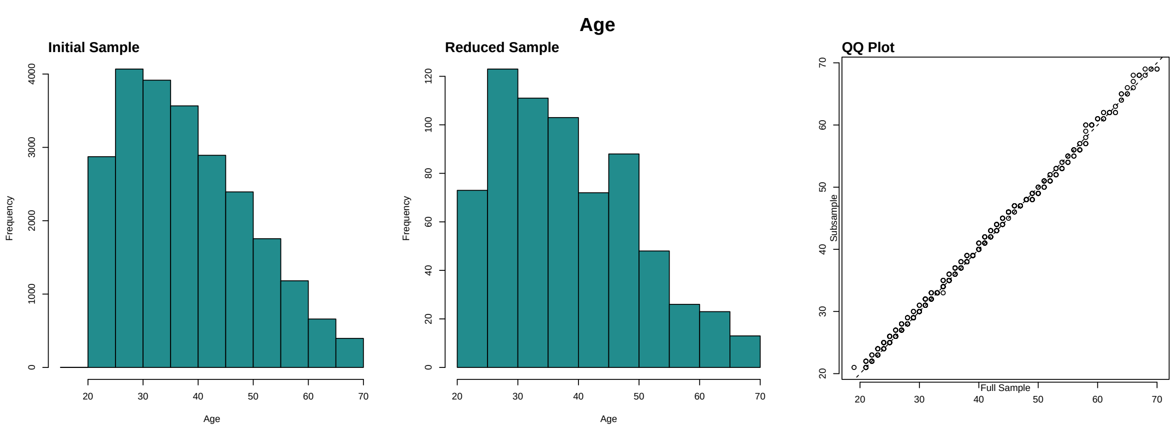


Figure 16: Distribution of Loan Duration across Samples

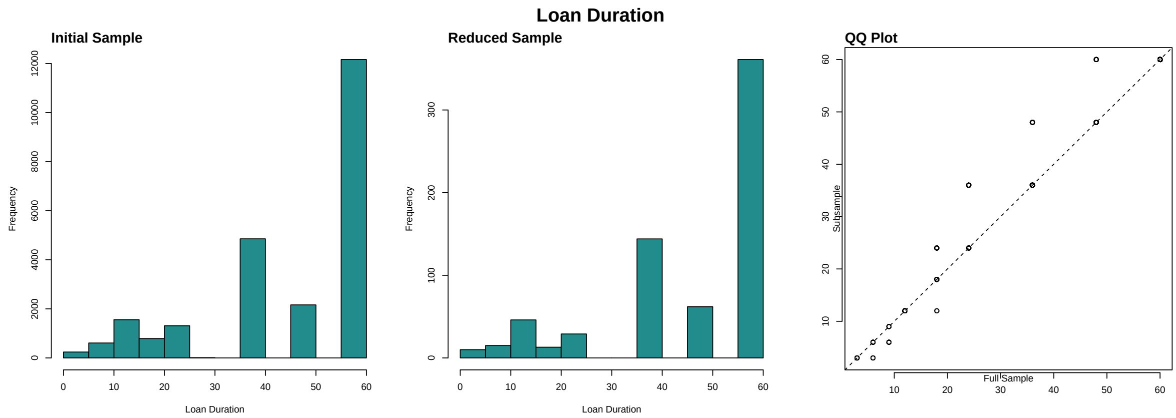


Figure 17: Distribution of Loan Purpose across Samples

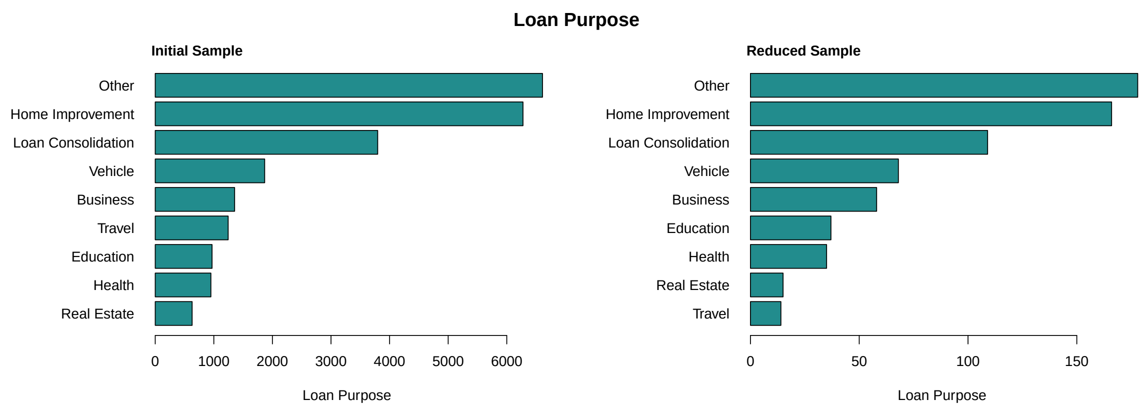


Figure 18: Distribution of Credit Ratings across Samples

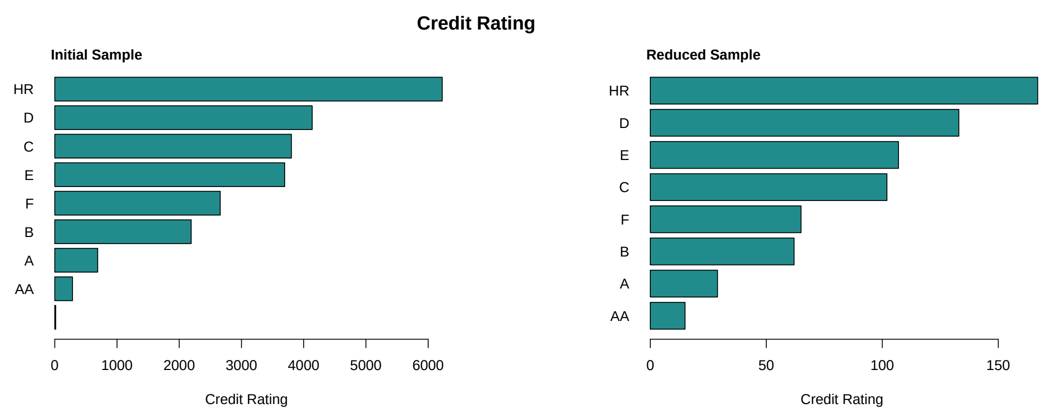


Figure 19: Distribution of Occupation Types across Samples

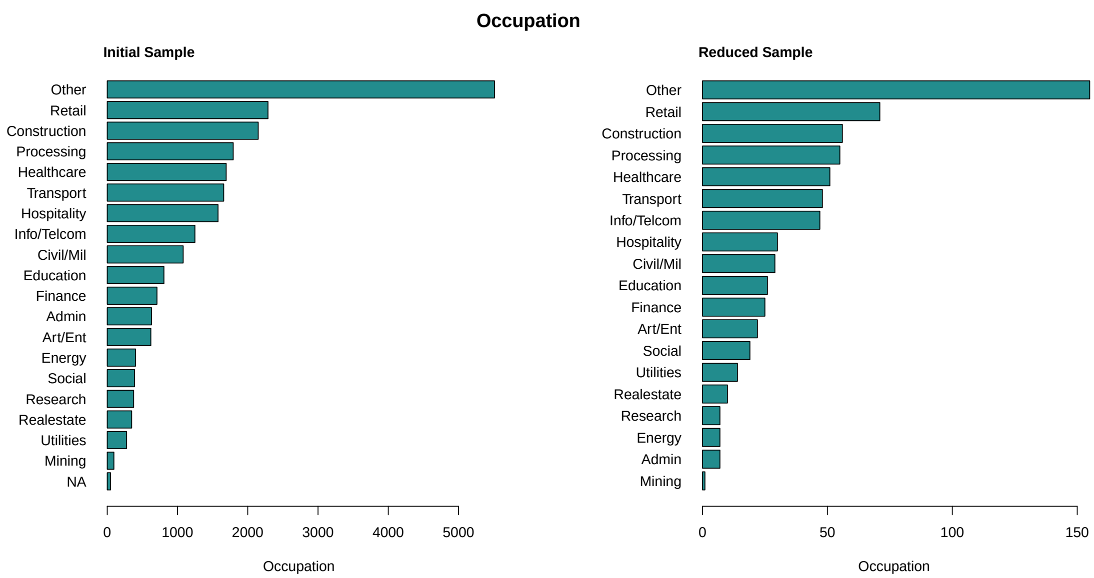
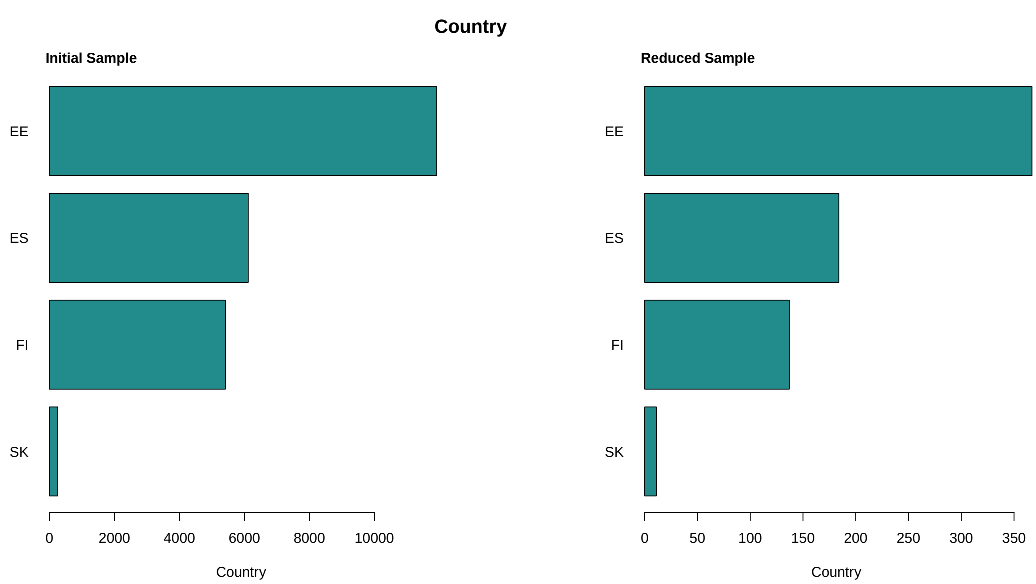


Figure 20: Distribution of Borrower Origins



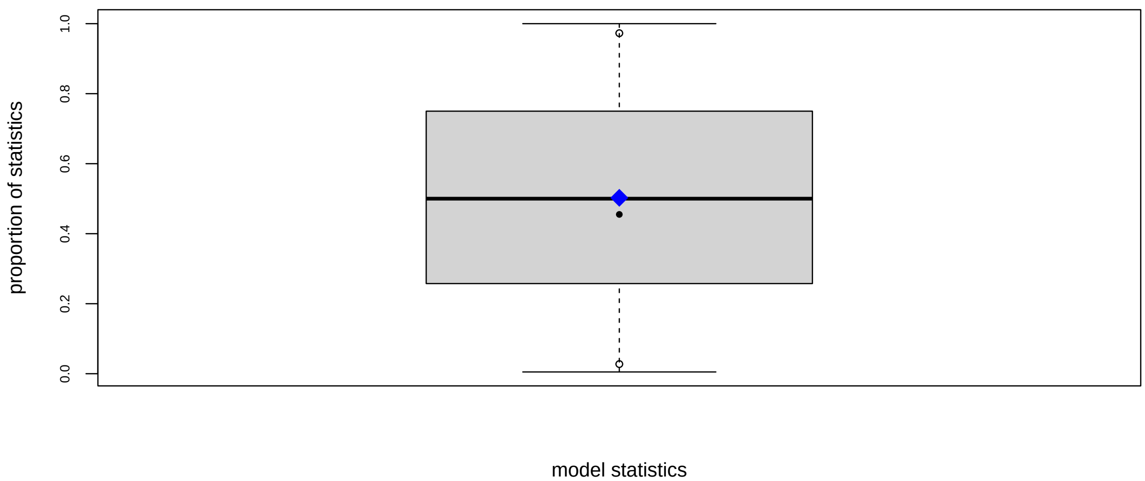
A.3 ERGM Results - Supplementary Tables

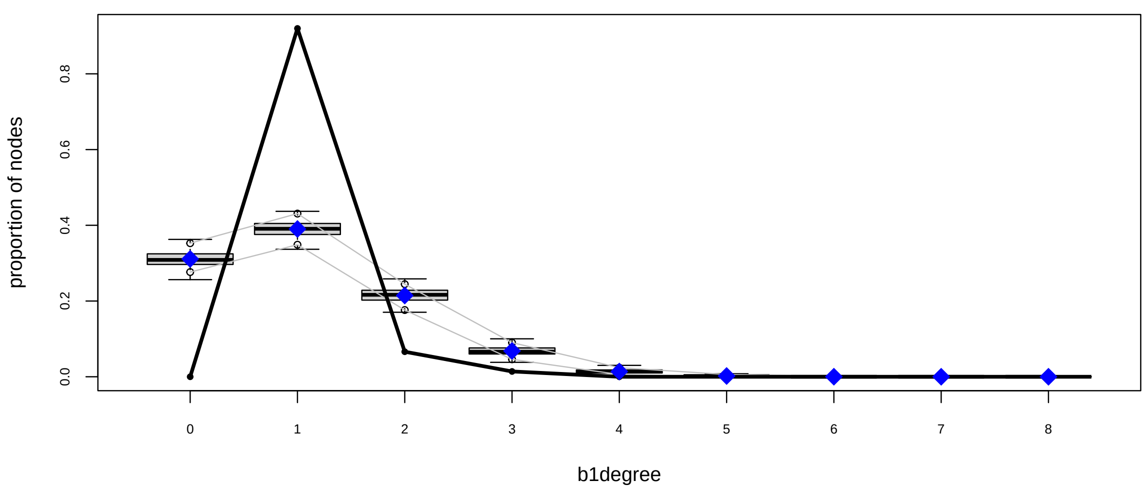
Table 6: ERGM Results in Probability Scale

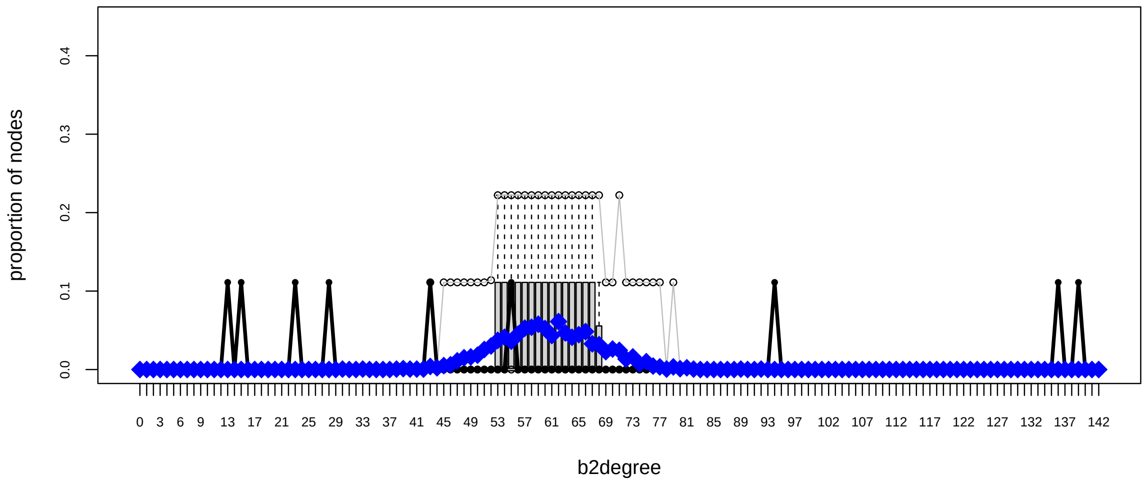
Terms	Base	b1degree.1.	cycle.4.	Age	Gender
edges	0.122	0.205	0.281	0.297	0.369
b1degree1	-	0.965	0.963	0.963	0.963
cycle4	-	-	0.465	0.465	0.463
b1factor.b1_gender.male	-	-	-	0.455	0.458
b1cov.b1_age	-	-	-	-	0.498

A.4 ERGM Diagnostics Plots

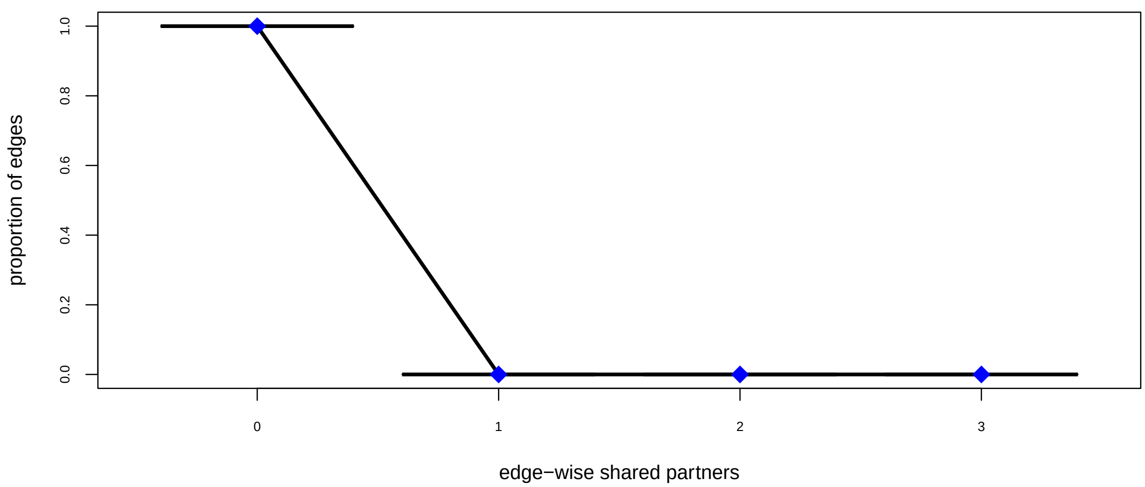
A.4.1 Base Model





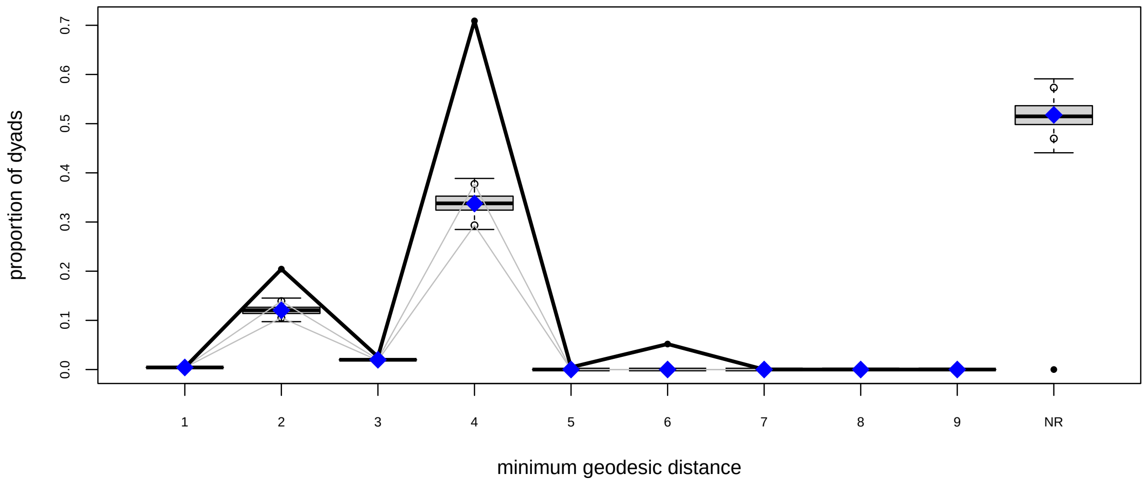


proportion of nodes
0.0
0.1
0.2
0.3
0.4
0 3 6 9 13 17 21 25 29 33 37 41 45 49 53 57 61 65 69 73 77 81 85 89 93 97 102 107 112 117 122 127 132 137 142
b2degree



proportion of edges
0.0
0.2
0.4
0.6
0.8
1.0
0
1
2
3
edge−wise shared partners

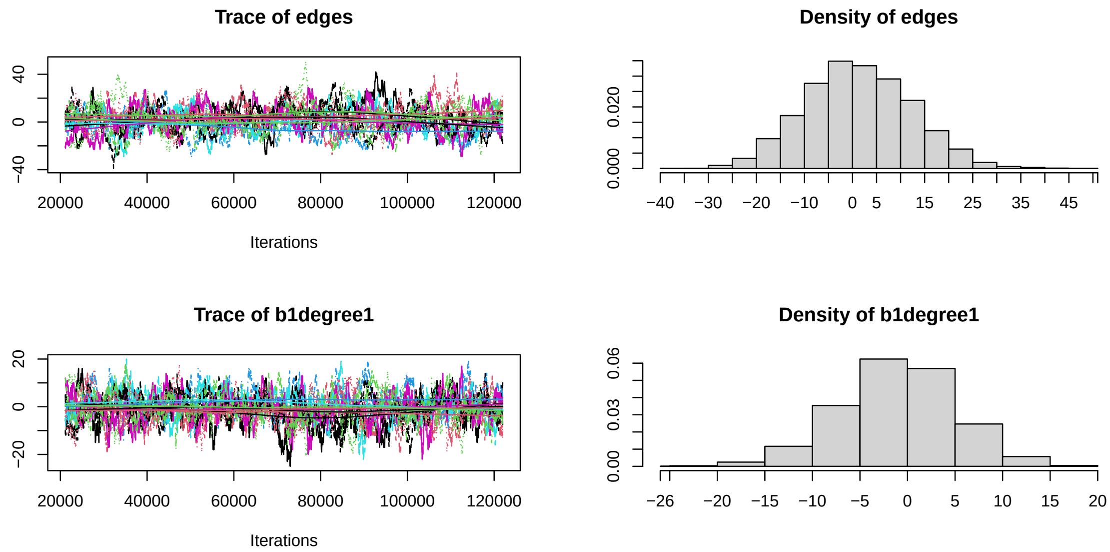
Goodness−of−fit diagnostics




```
Goodness-of-fit for model statistics

       obs        min       mean        max MC p-value
    546.00     487.00     543.86     602.00       0.91

Goodness-of-fit for minimum geodesic distance

      obs   min      mean   max MC p-value
1     546   487   543.860   602       0.91
2   26301 12531 15520.635 18711       0.00
3    3351  2348  2549.430  2749       0.00
4   91309 36680 43470.795 50047       0.00
5     594     0     0.460    48       0.00
6    6677     0     5.255   567       0.00
Inf     0 56768 66687.565 76128       0.00

Goodness-of-fit for bipartition 1 degree

  obs min    mean max MC p-value
0   0 128 155.250 183       0.00
1 459 168 194.705 230       0.00
2  33  73 107.065 129       0.00
3   7  19  34.040  50       0.00
4   0   2   6.865  15       0.00
5   0   0   1.005   4       0.75
6   0   0   0.070   1       1.00

Goodness-of-fit for bipartition 2 degree

    obs min  mean max MC p-value
13    1   0 0.000   0       0.00
15    1   0 0.000   0       0.00
23    1   0 0.000   0       0.00
28    1   0 0.000   0       0.00
30    0   0 0.005   1       1.00
33    0   0 0.005   1       1.00
38    0   0 0.005   1       1.00
39    0   0 0.010   1       1.00
40    0   0 0.005   1       1.00
41    0   0 0.005   1       1.00
42    0   0 0.005   1       1.00
43    1   0 0.035   1       0.07
```

```
44    0    0 0.020   1        1.00
45    0    0 0.045   1        1.00
46    0    0 0.055   1        1.00
47    0    0 0.100   1        1.00
48    0    0 0.135   2        1.00
49    0    0 0.145   2        1.00
50    0    0 0.165   2        1.00
51    0    0 0.230   2        1.00
52    0    0 0.270   2        1.00
53    0    0 0.335   3        1.00
54    0    0 0.370   3        1.00
55    1    0 0.325   2        0.56
56    0    0 0.405   3        1.00
57    0    0 0.475   3        1.00
58    0    0 0.485   2        1.00
59    0    0 0.520   3        1.00
60    0    0 0.470   3        1.00
61    0    0 0.390   2        1.00
62    0    0 0.550   3        1.00
63    0    0 0.420   3        1.00
64    0    0 0.370   3        1.00
65    0    0 0.400   3        1.00
66    0    0 0.435   4        1.00
67    0    0 0.295   3        1.00
68    0    0 0.285   3        1.00
69    0    0 0.205   2        1.00
70    0    0 0.235   2        1.00
71    0    0 0.220   2        1.00
72    0    0 0.120   2        1.00
73    0    0 0.145   3        1.00
74    0    0 0.065   1        1.00
75    0    0 0.090   1        1.00
76    0    0 0.040   1        1.00
77    0    0 0.030   1        1.00
78    0    0 0.005   1        1.00
79    0    0 0.030   1        1.00
80    0    0 0.010   1        1.00
81    0    0 0.020   1        1.00
82    0    0 0.005   1        1.00
89    0    0 0.005   1        1.00
92    0    0 0.005   1        1.00
94    1    0 0.000   0        0.00
136   1    0 0.000   0        0.00
139   1    0 0.000   0        0.00

Goodness-of-fit for edgewise shared partner

      obs        min       mean        max MC p-value
   546.00     487.00     543.86     602.00       0.91
```

A.4.2 Model 1

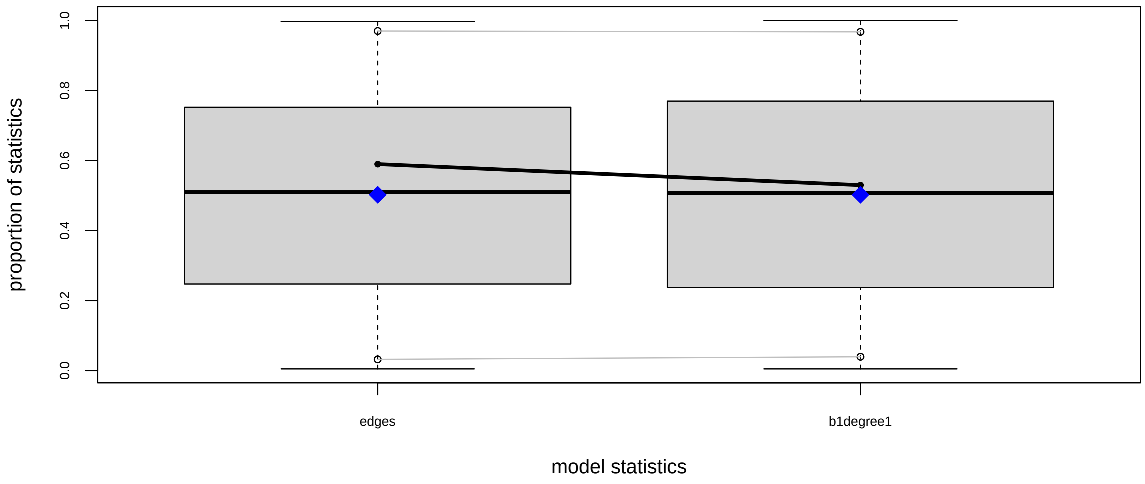


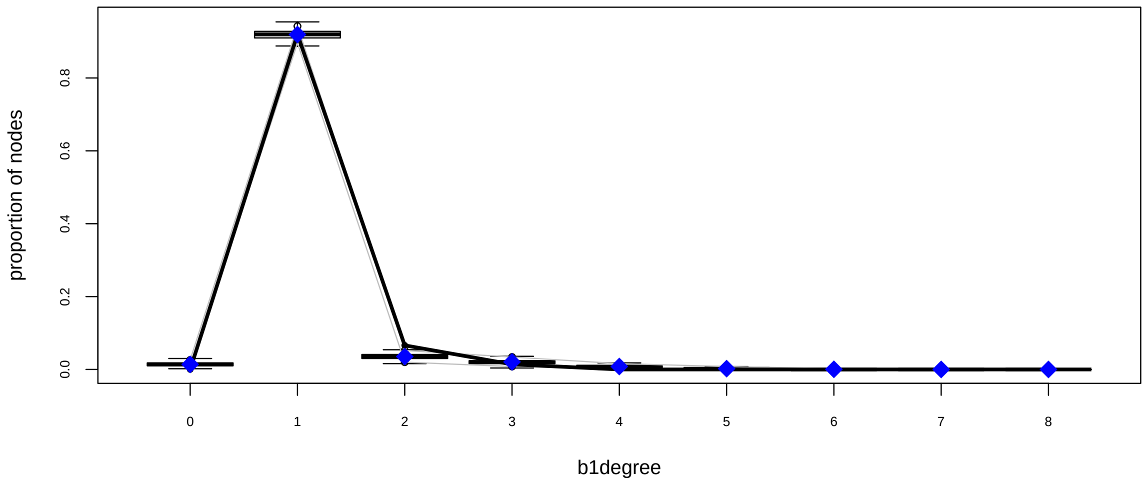

```
Sample statistics summary:

Iterations = 21136:121992
Thinning interval = 28
Number of chains = 9
Sample size per chain = 3603

1. Empirical mean and standard deviation for each variable,
   plus standard error of the mean:

            Mean     SD Naive SE Time-series SE
edges      1.727 11.066  0.06145         0.6592
b1degree1 -0.554  6.039  0.03353         0.2513

2. Quantiles for each variable:

          2.5% 25% 50% 75% 97.5%
edges      -19  -6   1   9    24
b1degree1  -13  -4   0   3    11


Are sample statistics significantly different from observed?
                 edges   b1degree1      (Omni)
diff.      1.726955932 -0.55404447          NA
test stat. 2.577412722 -2.16661869 5.80269221
P-val.     0.009954302  0.03026394 0.05651482

Sample statistics cross-correlations:
              edges  b1degree1
edges      1.0000000 -0.6736048
b1degree1 -0.6736048  1.0000000

Sample statistics auto-correlation:
Chain 1
            edges b1degree1
Lag 0   1.0000000 1.0000000
Lag 28  0.9826388 0.9701176
Lag 56  0.9664557 0.9439196
Lag 84  0.9498574 0.9180981
Lag 112 0.9335225 0.8938654
Lag 140 0.9173206 0.8721985
Chain 2
```

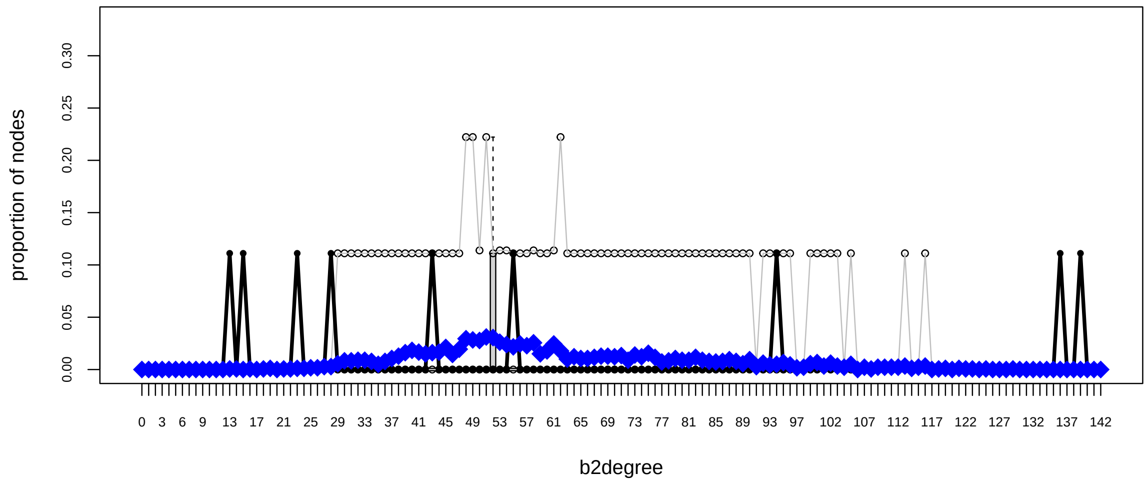
```
           edges b1degree1
Lag 0   1.0000000 1.0000000
Lag 28  0.9829174 0.9611738
Lag 56  0.9663965 0.9239480
Lag 84  0.9502249 0.8879717
Lag 112 0.9333931 0.8535116
Lag 140 0.9178077 0.8205518
Chain 3
           edges b1degree1
Lag 0   1.0000000 1.0000000
Lag 28  0.9883845 0.9692833
Lag 56  0.9772397 0.9397716
Lag 84  0.9662077 0.9126215
Lag 112 0.9556828 0.8871831
Lag 140 0.9453678 0.8626915
Chain 4
           edges b1degree1
Lag 0   1.0000000 1.0000000
Lag 28  0.9825875 0.9674543
Lag 56  0.9666999 0.9364074
Lag 84  0.9515609 0.9049248
Lag 112 0.9362933 0.8735428
Lag 140 0.9216203 0.8433670
Chain 5
           edges b1degree1
Lag 0   1.0000000 1.0000000
Lag 28  0.9817720 0.9637732
Lag 56  0.9642654 0.9307404
Lag 84  0.9471522 0.8990125
Lag 112 0.9306044 0.8686574
Lag 140 0.9141629 0.8394052
Chain 6
           edges b1degree1
Lag 0   1.0000000 1.0000000
Lag 28  0.9840958 0.9657033
Lag 56  0.9685792 0.9335799
Lag 84  0.9533266 0.9036673
Lag 112 0.9378353 0.8764645
Lag 140 0.9228011 0.8498216
Chain 7
           edges b1degree1
Lag 0   1.0000000 1.0000000
Lag 28  0.9852593 0.9626353
Lag 56  0.9707885 0.9284125
Lag 84  0.9574817 0.8962185
Lag 112 0.9445881 0.8654185
Lag 140 0.9318521 0.8363137
Chain 8
           edges b1degree1
Lag 0   1.0000000 1.0000000
Lag 28  0.9806314 0.9686263
Lag 56  0.9615573 0.9383344
Lag 84  0.9432512 0.9091995
Lag 112 0.9247775 0.8805019
Lag 140 0.9061377 0.8532768
Chain 9
           edges b1degree1
Lag 0   1.0000000 1.0000000
Lag 28  0.9795136 0.9606722
Lag 56  0.9603946 0.9238283
Lag 84  0.9417626 0.8881744
Lag 112 0.9231795 0.8544565
Lag 140 0.9036287 0.8209240

Sample statistics burn-in diagnostic (Geweke):
Chain 1

Fraction in 1st window = 0.1
```

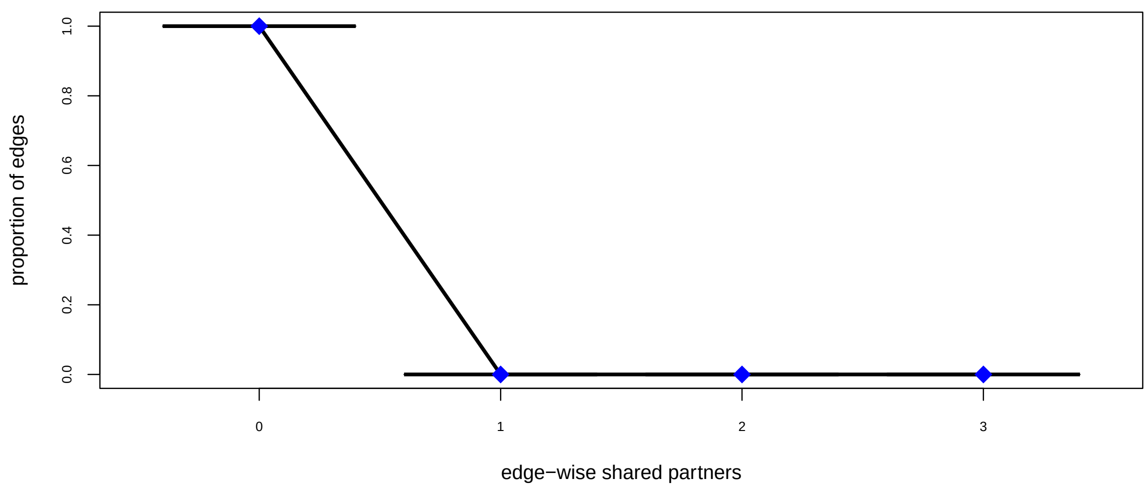
```
Fraction in 2nd window = 0.5

     edges  b1degree1
-0.8251148  1.1021806

Individual P-values (lower = worse):
    edges b1degree1
0.4093064 0.2703832
Joint P-value (lower = worse):  0.5497544
Chain 2

Fraction in 1st window = 0.1
Fraction in 2nd window = 0.5

     edges  b1degree1
-0.6130584  1.2345011

Individual P-values (lower = worse):
    edges b1degree1
0.5398377 0.2170163
Joint P-value (lower = worse):  0.3088801
Chain 3

Fraction in 1st window = 0.1
Fraction in 2nd window = 0.5

     edges  b1degree1
-0.5254825  1.8245035

Individual P-values (lower = worse):
     edges  b1degree1
0.59924780 0.06807598
Joint P-value (lower = worse):  0.3048263
Chain 4

Fraction in 1st window = 0.1
Fraction in 2nd window = 0.5

     edges  b1degree1
 2.1763745 -0.6342556

Individual P-values (lower = worse):
     edges  b1degree1
0.02952727 0.52591401
Joint P-value (lower = worse):  0.1381523
Chain 5

Fraction in 1st window = 0.1
Fraction in 2nd window = 0.5

      edges   b1degree1
 0.05826116 -0.14359472

Individual P-values (lower = worse):
    edges b1degree1
0.9535406 0.8858205
Joint P-value (lower = worse):  0.9955405
Chain 6

Fraction in 1st window = 0.1
Fraction in 2nd window = 0.5

     edges  b1degree1
-2.2393205  0.4044089

Individual P-values (lower = worse):
     edges  b1degree1
0.02513507 0.68591207
```

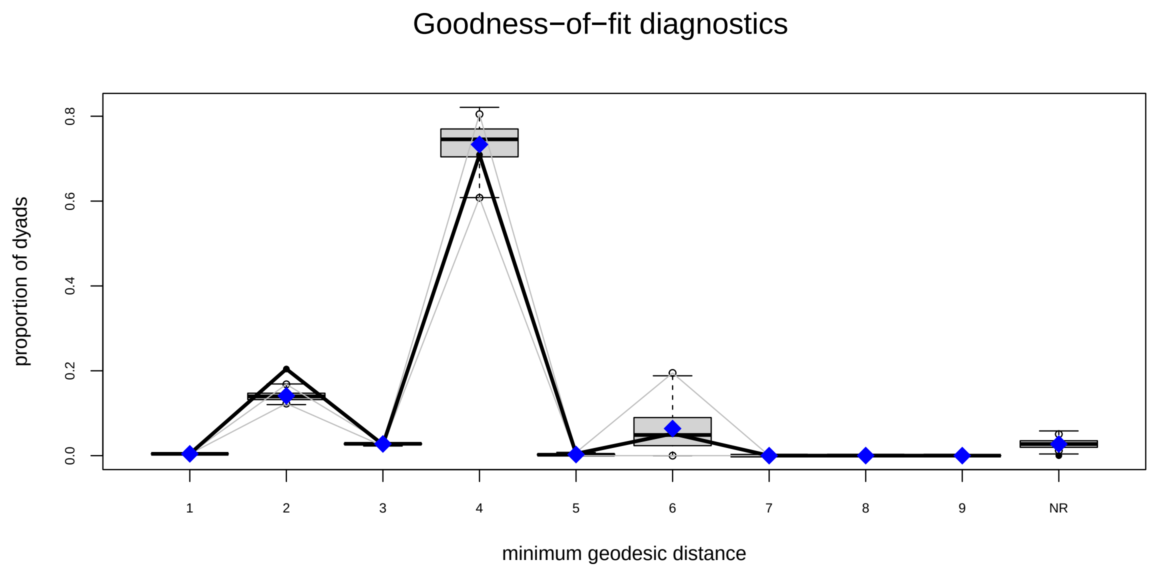
```
Joint P-value (lower = worse):  0.07543831
Chain 7

Fraction in 1st window = 0.1
Fraction in 2nd window = 0.5

     edges  b1degree1
 0.6404571 -0.4060245

Individual P-values (lower = worse):
    edges b1degree1
0.5218755 0.6847246
Joint P-value (lower = worse):  0.9039975
Chain 8

Fraction in 1st window = 0.1
Fraction in 2nd window = 0.5

     edges  b1degree1
-0.1195638 -0.2165394

Individual P-values (lower = worse):
    edges b1degree1
0.9048287 0.8285673
Joint P-value (lower = worse):  0.7900773
Chain 9

Fraction in 1st window = 0.1
Fraction in 2nd window = 0.5

    edges b1degree1
1.1789189 0.2317552

Individual P-values (lower = worse):
    edges b1degree1
0.2384305 0.8167281
Joint P-value (lower = worse):  0.4940679

Note: To save space, only one in every 7 iterations of the MCMC sample
  used for estimation was stored for diagnostics. Sample size per chain
  was originally around 25221 with thinning interval 4.

Note: MCMC diagnostics shown here are from the last round of
  simulation, prior to computation of final parameter estimates.
  Because the final estimates are refinements of those used for this
  simulation run, these diagnostics may understate model performance.
  To directly assess the performance of the final model on in-model
  statistics, please use the GOF command: gof(ergmFitObject,
  GOF=~model).
```


proportion of statistics
0.0
0.2
0.4
0.6
0.8
1.0
edges
b1degree1
model statistics


proportion of nodes
0.0
0.2
0.4
0.6
0.8
0
1
2
3
4
5
6
7
8
b1degree


proportion of nodes
0.00
0.05
0.10
0.15
0.20
0.25
0.30
0 3 6 9 13 17 21 25 29 33 37 41 45 49 53 57 61 65 69 73 77 81 85 89 93 97 102 107 112 117 122 127 132 137 142
b2degree


proportion of edges
0.0
0.2
0.4
0.6
0.8
1.0
0
1
2
3
edge−wise shared partners



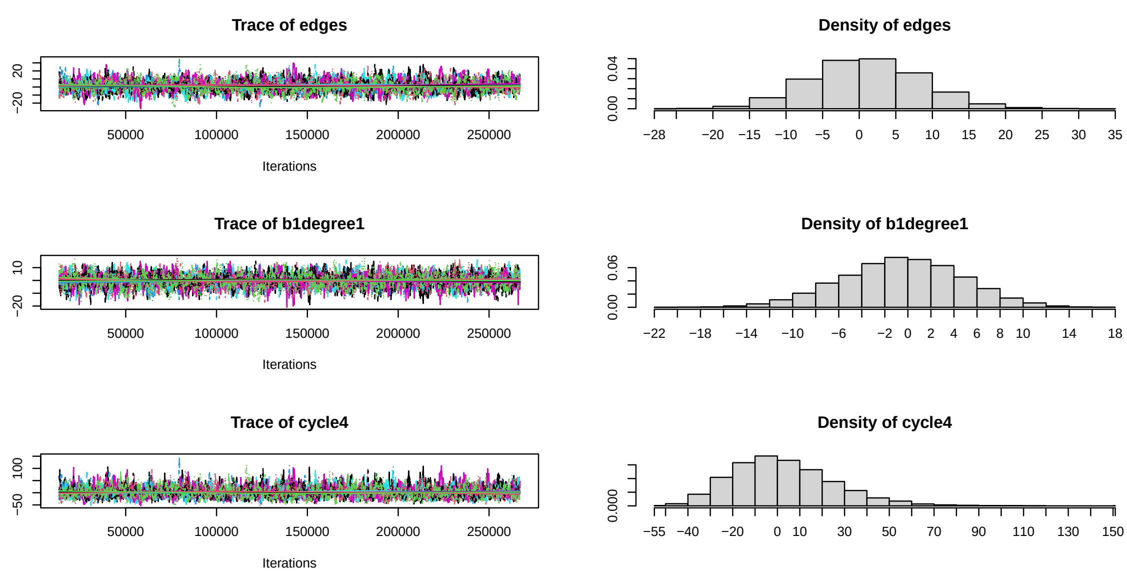

```
Goodness-of-fit for model statistics

          obs min    mean max MC p-value
edges     546 525 548.185 574       0.86
b1degree1 459 437 458.670 477       1.00

Goodness-of-fit for minimum geodesic distance

      obs   min      mean    max MC p-value
1     546   525   548.185    574       0.86
2   26301 15493 18153.770  23058       0.00
3    3351  2676  3547.225   3909       0.43
4   91309 74402 94453.820 105721       0.54
5     594     0   332.150   1251       0.32
6    6677     0  8222.595  31969       0.90
7       0     0     0.800    160       1.00
8       0     0    15.000   3000       1.00
Inf     0   507  3504.455   7992       0.00

Goodness-of-fit for bipartition 1 degree

  obs min    mean max MC p-value
0   0   1   6.960  16       0.00
1 459 437 458.670 477       1.00
2  33   8  17.630  31       0.00
3   7   2  10.340  24       0.39
4   0   0   4.040  10       0.05
5   0   0   1.120   4       0.68
6   0   0   0.210   2       1.00
7   0   0   0.025   2       1.00
8   0   0   0.005   1       1.00

Goodness-of-fit for bipartition 2 degree

    obs min  mean max MC p-value
13    1   0 0.005   1       0.01
14    0   0 0.005   1       1.00
15    1   0 0.000   0       0.00
16    0   0 0.005   1       1.00
18    0   0 0.005   1       1.00
19    0   0 0.010   1       1.00
21    0   0 0.005   1       1.00
```

22	0	0	0.005	1	1.00
23	1	0	0.010	1	0.02
24	0	0	0.010	1	1.00
25	0	0	0.015	1	1.00
26	0	0	0.015	1	1.00
27	0	0	0.025	2	1.00
28	1	0	0.025	1	0.05
29	0	0	0.050	2	1.00
30	0	0	0.075	1	1.00
31	0	0	0.075	1	1.00
32	0	0	0.080	2	1.00
33	0	0	0.080	1	1.00
34	0	0	0.070	1	1.00
35	0	0	0.045	1	1.00
36	0	0	0.070	1	1.00
37	0	0	0.095	2	1.00
38	0	0	0.115	1	1.00
39	0	0	0.145	2	1.00
40	0	0	0.165	2	1.00
41	0	0	0.150	2	1.00
42	0	0	0.135	1	1.00
43	1	0	0.145	2	0.27
44	0	0	0.150	1	1.00
45	0	0	0.190	2	1.00
46	0	0	0.130	2	1.00
47	0	0	0.175	2	1.00
48	0	0	0.265	3	1.00
49	0	0	0.255	2	1.00
50	0	0	0.250	2	1.00
51	0	0	0.280	3	1.00
52	0	0	0.275	2	1.00
53	0	0	0.235	2	1.00
54	0	0	0.215	2	1.00
55	1	0	0.195	2	0.38
56	0	0	0.220	2	1.00
57	0	0	0.205	2	1.00
58	0	0	0.230	3	1.00
59	0	0	0.135	2	1.00
60	0	0	0.160	2	1.00
61	0	0	0.220	3	1.00
62	0	0	0.160	2	1.00
63	0	0	0.090	2	1.00
64	0	0	0.110	2	1.00
65	0	0	0.095	2	1.00
66	0	0	0.095	2	1.00
67	0	0	0.105	2	1.00
68	0	0	0.115	1	1.00
69	0	0	0.115	2	1.00
70	0	0	0.110	3	1.00
71	0	0	0.120	2	1.00
72	0	0	0.080	1	1.00
73	0	0	0.125	1	1.00
74	0	0	0.110	2	1.00
75	0	0	0.140	2	1.00
76	0	0	0.105	1	1.00
77	0	0	0.065	1	1.00
78	0	0	0.075	1	1.00
79	0	0	0.095	2	1.00
80	0	0	0.080	2	1.00
81	0	0	0.080	1	1.00
82	0	0	0.105	2	1.00
83	0	0	0.080	1	1.00
84	0	0	0.070	1	1.00
85	0	0	0.065	1	1.00
86	0	0	0.070	1	1.00
87	0	0	0.085	2	1.00
88	0	0	0.070	1	1.00
89	0	0	0.050	2	1.00

```
90     0    0 0.085   1          1.00
91     0    0 0.025   1          1.00
92     0    0 0.055   2          1.00
93     0    0 0.040   2          1.00
94     1    0 0.045   1          0.09
95     0    0 0.060   1          1.00
96     0    0 0.040   1          1.00
97     0    0 0.015   1          1.00
98     0    0 0.020   1          1.00
99     0    0 0.050   1          1.00
100    0    0 0.060   2          1.00
101    0    0 0.030   1          1.00
102    0    0 0.055   1          1.00
103    0    0 0.030   1          1.00
104    0    0 0.020   2          1.00
105    0    0 0.045   1          1.00
107    0    0 0.020   1          1.00
108    0    0 0.005   1          1.00
109    0    0 0.020   2          1.00
110    0    0 0.020   1          1.00
111    0    0 0.020   1          1.00
112    0    0 0.020   1          1.00
113    0    0 0.030   1          1.00
114    0    0 0.010   1          1.00
115    0    0 0.020   1          1.00
116    0    0 0.030   1          1.00
118    0    0 0.005   1          1.00
119    0    0 0.010   1          1.00
121    0    0 0.010   1          1.00
122    0    0 0.005   1          1.00
123    0    0 0.005   1          1.00
125    0    0 0.005   1          1.00
129    0    0 0.005   1          1.00
136    1    0 0.000   0          0.00
139    1    0 0.000   0          0.00
```

```
Goodness-of-fit for edgewise shared partner

      obs        min       mean        max MC p-value
  546.000    525.000    548.185    574.000      0.860
```

A.4.3 Model 2

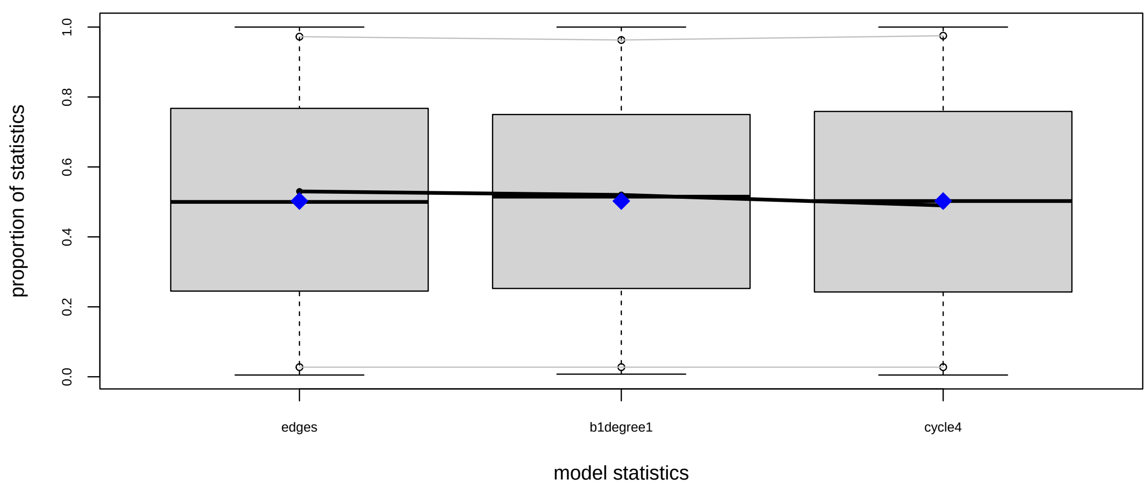


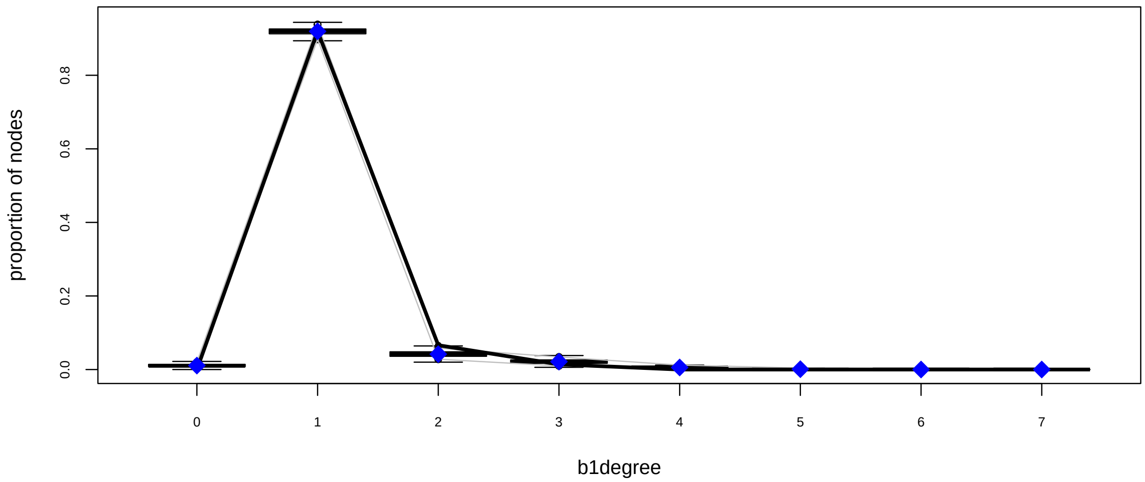
```
Sample statistics summary:

Iterations = 13376:267072
Thinning interval = 64
Number of chains = 9
Sample size per chain = 3965

1. Empirical mean and standard deviation for each variable,
   plus standard error of the mean:

             Mean     SD Naive SE Time-series SE
edges      1.44649  7.475  0.03957         0.1970
b1degree1 -0.05955  5.224  0.02766         0.1202
cycle4     3.02365 22.559  0.11942         0.5202

2. Quantiles for each variable:

          2.5% 25% 50% 75% 97.5%
edges      -13  -4   1   7    16
b1degree1  -10  -4   0   4    10
cycle4     -34 -13   1  17    53


Are sample statistics significantly different from observed?
                 edges   b1degree1       cycle4       (Omni)
diff.      1.446490e+00 -0.05954883 3.023651e+00           NA
test stat. 7.380226e+00 -0.49639643 5.879148e+00 8.382182e+01
P-val.     1.580205e-13  0.61961475 4.123823e-09 1.072914e-17

Sample statistics cross-correlations:
              edges b1degree1     cycle4
edges      1.0000000 -0.644952  0.8593833
b1degree1 -0.6449520  1.000000 -0.5739620
cycle4     0.8593833 -0.573962  1.0000000

Sample statistics auto-correlation:
Chain 1
           edges b1degree1    cycle4
Lag 0   1.0000000 1.0000000 1.0000000
Lag 64  0.9207720 0.8939305 0.8908135
Lag 128 0.8477613 0.7974903 0.7902465
Lag 192 0.7824516 0.7176731 0.7027372
Lag 256 0.7199936 0.6447281 0.6311525
Lag 320 0.6593805 0.5796761 0.5623123
Chain 2
           edges b1degree1    cycle4
Lag 0   1.0000000 1.0000000 1.0000000
Lag 64  0.9127709 0.8916141 0.8699662
Lag 128 0.8323720 0.7990969 0.7576879
Lag 192 0.7585718 0.7152578 0.6632909
Lag 256 0.6928837 0.6408781 0.5816583
Lag 320 0.6307202 0.5757939 0.5101108
Chain 3
           edges b1degree1    cycle4
Lag 0   1.0000000 1.0000000 1.0000000
Lag 64  0.9247032 0.9053829 0.8947725
Lag 128 0.8626373 0.8250421 0.8072021
Lag 192 0.8036997 0.7535264 0.7285900
Lag 256 0.7514687 0.6915726 0.6666843
Lag 320 0.7033289 0.6395590 0.6111047
Chain 4
           edges b1degree1    cycle4
Lag 0   1.0000000 1.0000000 1.0000000
Lag 64  0.9194227 0.8882619 0.8836353
Lag 128 0.8460630 0.7872335 0.7865274
Lag 192 0.7760913 0.6949317 0.7035127
Lag 256 0.7112373 0.6160141 0.6286306
Lag 320 0.6561840 0.5565084 0.5674450
```

```
Chain 5
            edges b1degree1    cycle4
Lag 0   1.0000000 1.0000000 1.0000000
Lag 64  0.9216747 0.8899834 0.8896689
Lag 128 0.8516064 0.7949439 0.7975814
Lag 192 0.7891529 0.7120355 0.7220413
Lag 256 0.7350742 0.6384775 0.6582887
Lag 320 0.6902214 0.5758803 0.6053976
Chain 6
            edges b1degree1    cycle4
Lag 0   1.0000000 1.0000000 1.0000000
Lag 64  0.9286170 0.9040554 0.8938356
Lag 128 0.8635580 0.8200027 0.8061285
Lag 192 0.8054181 0.7433293 0.7345318
Lag 256 0.7518688 0.6798884 0.6733401
Lag 320 0.7022343 0.6197807 0.6207549
Chain 7
            edges b1degree1    cycle4
Lag 0   1.0000000 1.0000000 1.0000000
Lag 64  0.9146513 0.8928942 0.8829731
Lag 128 0.8378383 0.7951906 0.7895102
Lag 192 0.7709455 0.7122812 0.7088251
Lag 256 0.7106735 0.6369230 0.6390087
Lag 320 0.6506091 0.5680243 0.5719387
Chain 8
            edges b1degree1    cycle4
Lag 0   1.0000000 1.0000000 1.0000000
Lag 64  0.9169727 0.8948412 0.8835476
Lag 128 0.8384772 0.7985448 0.7808854
Lag 192 0.7696477 0.7101403 0.6929739
Lag 256 0.7056378 0.6336532 0.6158042
Lag 320 0.6465219 0.5651076 0.5422954
Chain 9
            edges b1degree1    cycle4
Lag 0   1.0000000 1.0000000 1.0000000
Lag 64  0.9153677 0.8944287 0.8843268
Lag 128 0.8355754 0.7943154 0.7820059
Lag 192 0.7669759 0.7137630 0.6998537
Lag 256 0.7064500 0.6496772 0.6291565
Lag 320 0.6520078 0.5914224 0.5688262

Sample statistics burn-in diagnostic (Geweke):
Chain 1

Fraction in 1st window = 0.1
Fraction in 2nd window = 0.5

     edges  b1degree1     cycle4
-0.9336618 -0.3072449  0.1164942

Individual P-values (lower = worse):
    edges b1degree1    cycle4
0.3504784 0.7586570 0.9072609
Joint P-value (lower = worse):  0.1702401
Chain 2

Fraction in 1st window = 0.1
Fraction in 2nd window = 0.5

       edges    b1degree1       cycle4
 0.009917489 -0.849595199  0.751238982

Individual P-values (lower = worse):
    edges b1degree1    cycle4
0.9920871 0.3955502 0.4525088
Joint P-value (lower = worse):  0.4049945
Chain 3
```

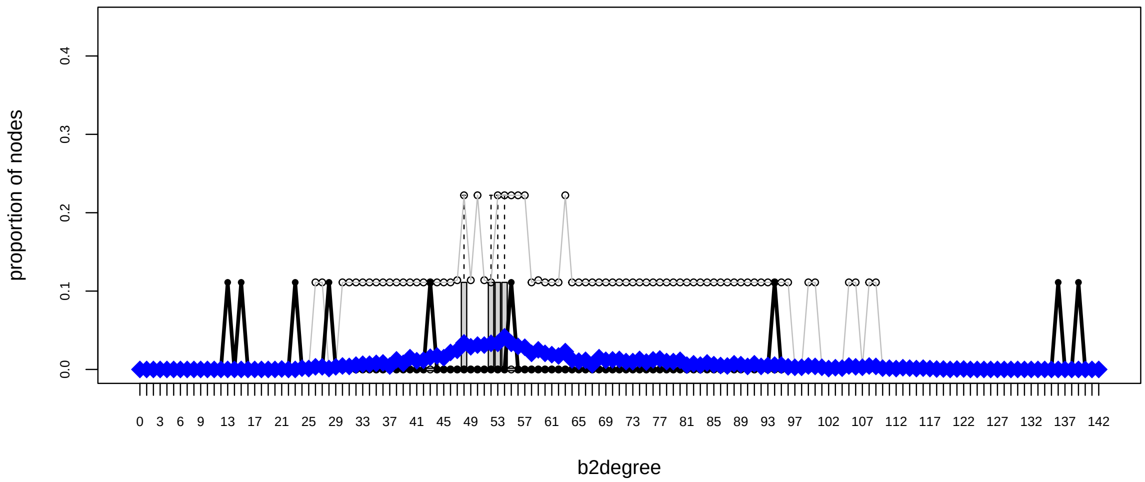
```
Fraction in 1st window = 0.1
Fraction in 2nd window = 0.5

     edges  b1degree1     cycle4
-0.2985873  0.6818755 -1.1698052

Individual P-values (lower = worse):
    edges b1degree1    cycle4
0.7652550 0.4953177 0.2420794
Joint P-value (lower = worse):  0.5700744
Chain 4

Fraction in 1st window = 0.1
Fraction in 2nd window = 0.5

     edges  b1degree1     cycle4
 0.1115741 -0.9752093  0.1768475

Individual P-values (lower = worse):
    edges b1degree1    cycle4
0.9111611 0.3294564 0.8596282
Joint P-value (lower = worse):  0.5713822
Chain 5

Fraction in 1st window = 0.1
Fraction in 2nd window = 0.5

     edges  b1degree1     cycle4
-0.1818034  0.3848732 -0.4954972

Individual P-values (lower = worse):
    edges b1degree1    cycle4
0.8557370 0.7003313 0.6202492
Joint P-value (lower = worse):  0.8975539
Chain 6

Fraction in 1st window = 0.1
Fraction in 2nd window = 0.5

     edges  b1degree1     cycle4
-0.4751854  0.8079716 -0.2877042

Individual P-values (lower = worse):
    edges b1degree1    cycle4
0.6346548 0.4191069 0.7735732
Joint P-value (lower = worse):  0.6088046
Chain 7

Fraction in 1st window = 0.1
Fraction in 2nd window = 0.5

     edges  b1degree1     cycle4
-0.8753117 -0.1067233 -0.7792244

Individual P-values (lower = worse):
    edges b1degree1    cycle4
0.3814043 0.9150085 0.4358475
Joint P-value (lower = worse):  0.5840519
Chain 8

Fraction in 1st window = 0.1
Fraction in 2nd window = 0.5

     edges  b1degree1     cycle4
-0.4947983  1.4777652 -1.9407711

Individual P-values (lower = worse):
     edges  b1degree1     cycle4
```

```
0.62074253 0.13947063 0.05228605
Joint P-value (lower = worse):  0.1057777
Chain 9

Fraction in 1st window = 0.1
Fraction in 2nd window = 0.5

    edges b1degree1    cycle4
0.7249734 0.1051507 0.8464796

Individual P-values (lower = worse):
    edges b1degree1    cycle4
0.4684684 0.9162563 0.3972853
Joint P-value (lower = worse):  0.725172

Note: To save space, only one in every 4 iterations of the MCMC sample
  used for estimation was stored for diagnostics. Sample size per chain
  was originally around 15860 with thinning interval 16.

Note: MCMC diagnostics shown here are from the last round of
  simulation, prior to computation of final parameter estimates.
  Because the final estimates are refinements of those used for this
  simulation run, these diagnostics may understate model performance.
  To directly assess the performance of the final model on in-model
  statistics, please use the GOF command: gof(ergmFitObject,
  GOF=~model).
```

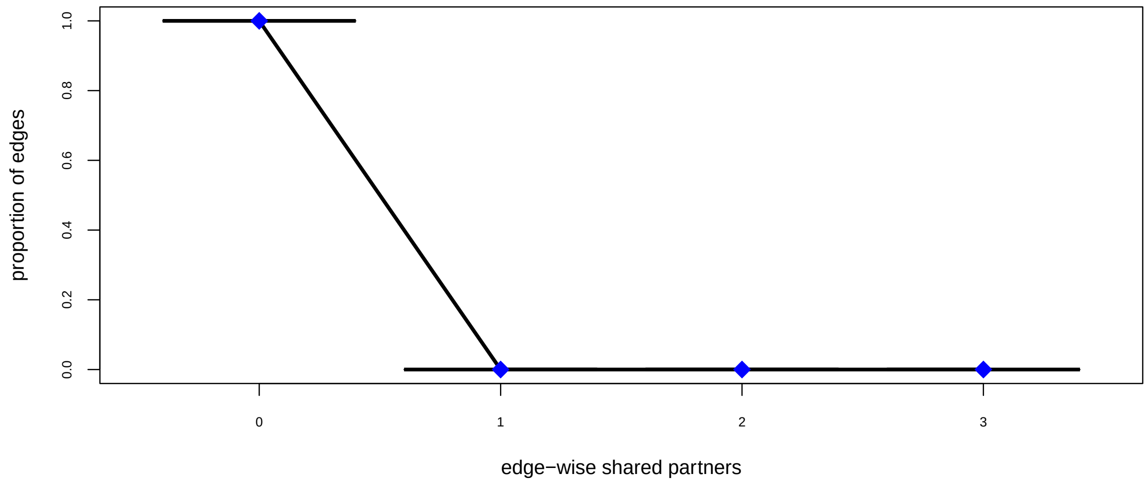




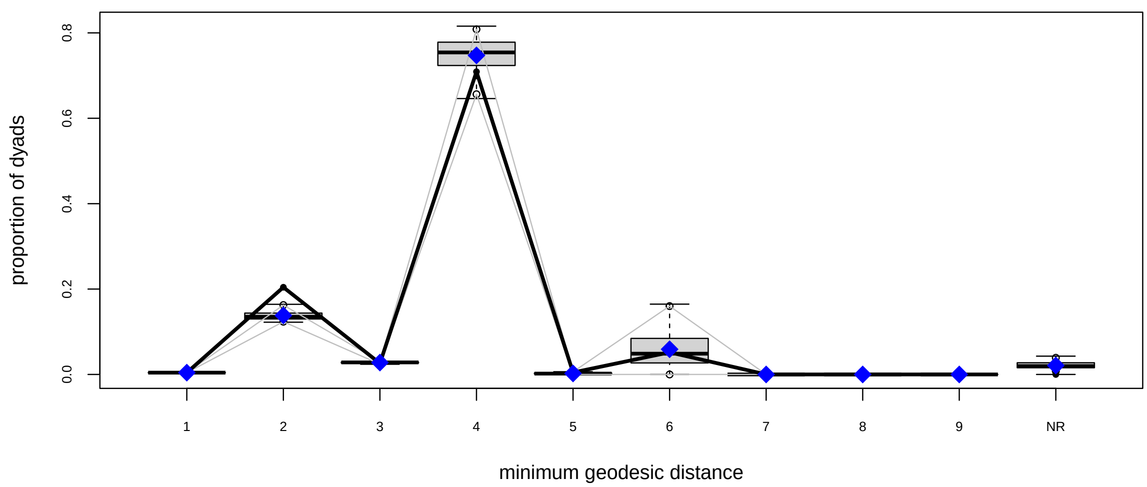
proportion of nodes
0.0
0.2
0.4
0.6
0.8
0
1
2
3
4
5
6
7
b1degree



proportion of nodes
0.0
0.1
0.2
0.3
0.4
0 3 6 9 13 17 21 25 29 33 37 41 45 49 53 57 61 65 69 73 77 81 85 89 93 97 102 107 112 117 122 127 132 137 142
b2degree



Goodness−of−fit diagnostics



Goodness-of-fit for model statistics

	obs	min	mean	max	MC p-value
edges	546	528	546.330	568	1.00
b1degree1	459	446	458.785	473	1.00
cycle4	67	26	68.885	157	0.98

Goodness-of-fit for minimum geodesic distance

	obs	min	mean	max	MC p-value
1	546	528	546.330	568	1.00
2	26301	15759	17794.830	21669	0.00
3	3351	2845	3587.910	3920	0.33
4	91309	77582	96233.180	105053	0.37
5	594	0	308.520	1061	0.25
6	6677	0	7603.865	28137	0.96
Inf	0	0	2703.365	6018	0.01

Goodness-of-fit for bipartition 1 degree

	obs	min	mean	max	MC p-value
0	0	0	5.360	12	0.01
1	459	446	458.785	473	1.00
2	33	10	20.880	32	0.00
3	7	3	10.670	21	0.30
4	0	0	2.790	7	0.10
5	0	0	0.475	3	1.00
6	0	0	0.040	2	1.00

Goodness-of-fit for bipartition 2 degree

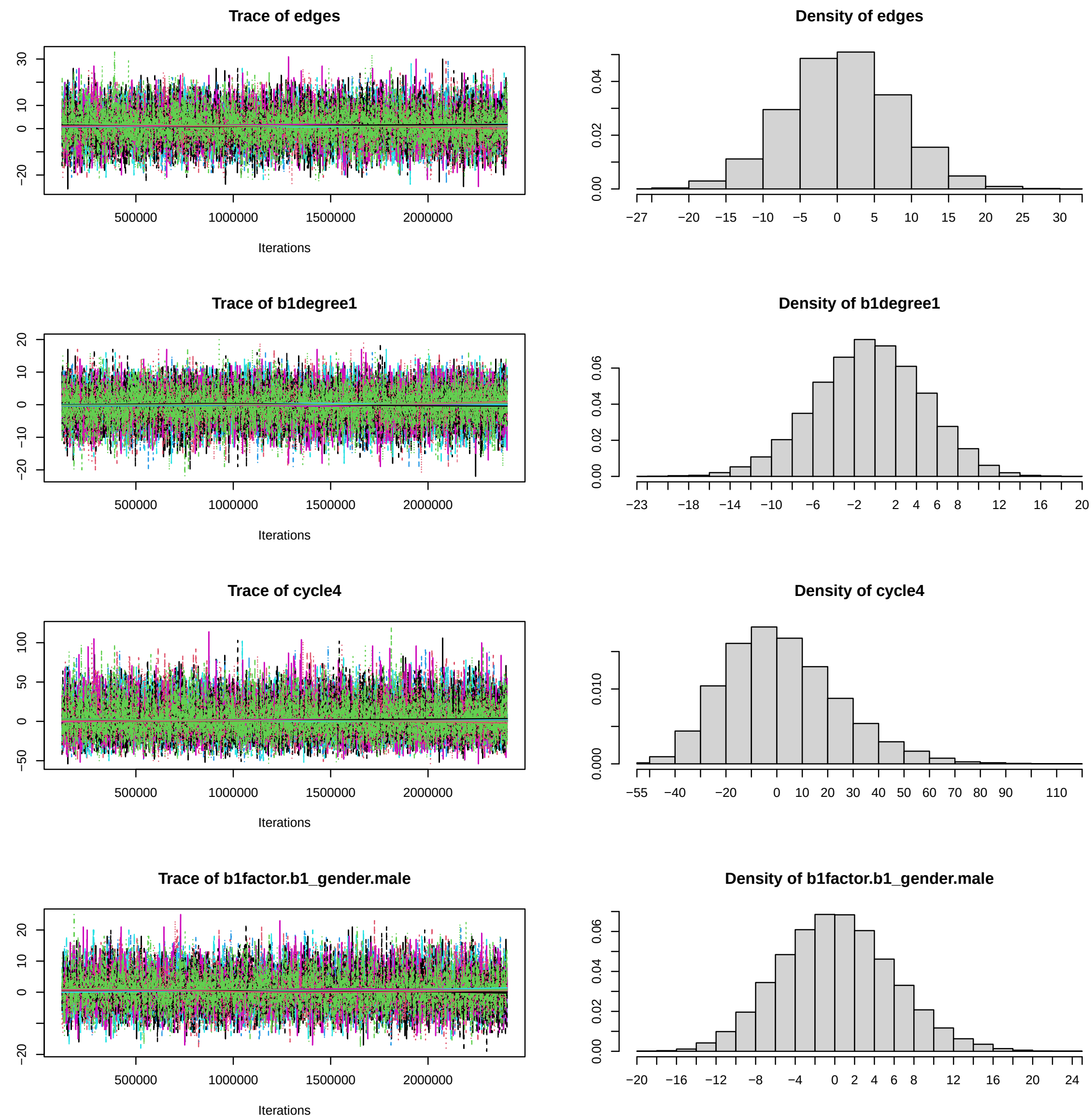
	obs	min	mean	max	MC p-value
13	1	0	0.000	0	0.00
15	1	0	0.000	0	0.00
21	0	0	0.005	1	1.00
23	1	0	0.000	0	0.00
24	0	0	0.015	2	1.00
25	0	0	0.010	1	1.00
26	0	0	0.030	1	1.00
27	0	0	0.030	1	1.00
28	1	0	0.010	1	0.02
29	0	0	0.030	2	1.00
30	0	0	0.040	2	1.00
31	0	0	0.035	1	1.00
32	0	0	0.050	2	1.00
33	0	0	0.060	1	1.00
34	0	0	0.060	1	1.00
35	0	0	0.070	2	1.00
36	0	0	0.075	1	1.00
37	0	0	0.040	1	1.00
38	0	0	0.110	1	1.00
39	0	0	0.070	2	1.00
40	0	0	0.135	1	1.00
41	0	0	0.100	3	1.00
42	0	0	0.105	3	1.00
43	1	0	0.140	2	0.27
44	0	0	0.155	2	1.00
45	0	0	0.135	2	1.00
46	0	0	0.195	2	1.00
47	0	0	0.220	2	1.00
48	0	0	0.305	2	1.00
49	0	0	0.260	3	1.00
50	0	0	0.280	3	1.00
51	0	0	0.280	3	1.00
52	0	0	0.300	2	1.00
53	0	0	0.300	2	1.00
54	0	0	0.375	2	1.00
55	1	0	0.300	3	0.48
56	0	0	0.265	4	1.00
57	0	0	0.255	2	1.00
58	0	0	0.185	3	1.00
59	0	0	0.225	2	1.00
60	0	0	0.185	3	1.00
61	0	0	0.170	2	1.00
62	0	0	0.155	3	1.00
63	0	0	0.205	3	1.00
64	0	0	0.110	2	1.00
65	0	0	0.095	1	1.00
66	0	0	0.105	2	1.00
67	0	0	0.055	2	1.00
68	0	0	0.135	2	1.00
69	0	0	0.105	2	1.00
70	0	0	0.110	2	1.00
71	0	0	0.115	2	1.00
72	0	0	0.090	2	1.00

73	0	0	0.090	2	1.00
74	0	0	0.115	2	1.00
75	0	0	0.085	2	1.00
76	0	0	0.110	2	1.00
77	0	0	0.120	1	1.00
78	0	0	0.090	2	1.00
79	0	0	0.090	2	1.00
80	0	0	0.105	2	1.00
81	0	0	0.050	1	1.00
82	0	0	0.065	1	1.00
83	0	0	0.050	1	1.00
84	0	0	0.075	1	1.00
85	0	0	0.055	1	1.00
86	0	0	0.045	1	1.00
87	0	0	0.040	2	1.00
88	0	0	0.065	2	1.00
89	0	0	0.055	1	1.00
90	0	0	0.035	1	1.00
91	0	0	0.065	1	1.00
92	0	0	0.035	2	1.00
93	0	0	0.045	1	1.00
94	1	0	0.050	2	0.09
95	0	0	0.050	1	1.00
96	0	0	0.030	1	1.00
97	0	0	0.025	1	1.00
98	0	0	0.025	1	1.00
99	0	0	0.040	1	1.00
100	0	0	0.035	1	1.00
101	0	0	0.025	1	1.00
102	0	0	0.010	1	1.00
103	0	0	0.020	1	1.00
104	0	0	0.015	1	1.00
105	0	0	0.040	1	1.00
106	0	0	0.030	1	1.00
107	0	0	0.025	1	1.00
108	0	0	0.040	1	1.00
109	0	0	0.035	1	1.00
110	0	0	0.015	1	1.00
111	0	0	0.015	1	1.00
112	0	0	0.010	1	1.00
113	0	0	0.020	1	1.00
114	0	0	0.015	1	1.00
115	0	0	0.010	1	1.00
116	0	0	0.015	1	1.00
117	0	0	0.010	1	1.00
118	0	0	0.005	1	1.00
119	0	0	0.005	1	1.00
121	0	0	0.005	1	1.00
122	0	0	0.005	1	1.00
136	1	0	0.000	0	0.00
139	1	0	0.000	0	0.00

Goodness-of-fit for edgewise shared partner

obs	min	mean	max	MC p-value
546.00	528.00	546.33	568.00	1.00

A.4.4 Model 3



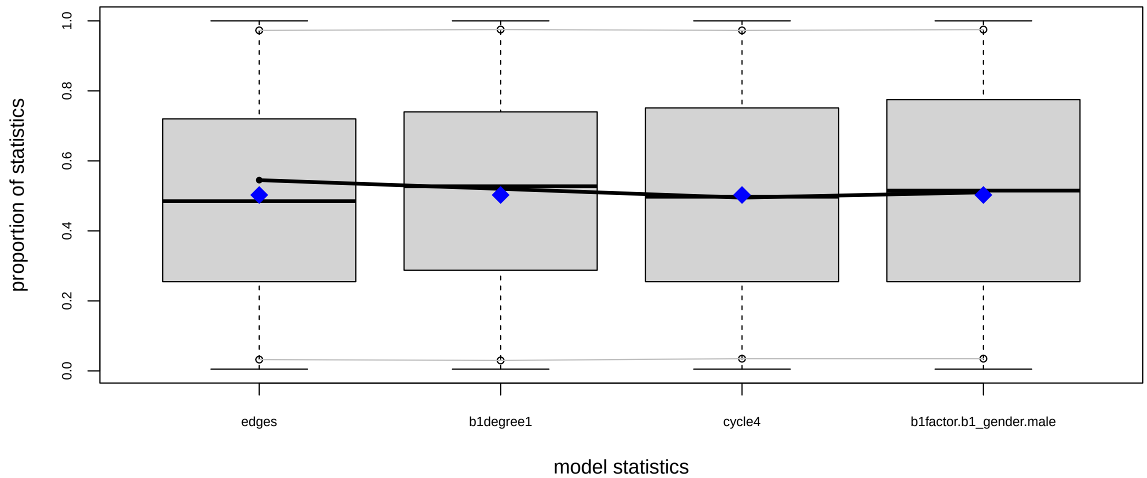

```
Sample statistics summary:

Iterations = 120448:2406528
Thinning interval = 640
Number of chains = 9
Sample size per chain = 3573

1. Empirical mean and standard deviation for each variable,
   plus standard error of the mean:

                            Mean     SD Naive SE Time-series SE
edges                    1.25307  7.431  0.04144        0.06826
b1degree1               -0.05458  5.252  0.02929        0.04472
cycle4                   2.61648 22.356  0.12467        0.18772
b1factor.b1_gender.male  0.71207  5.713  0.03186        0.05854

2. Quantiles for each variable:
```

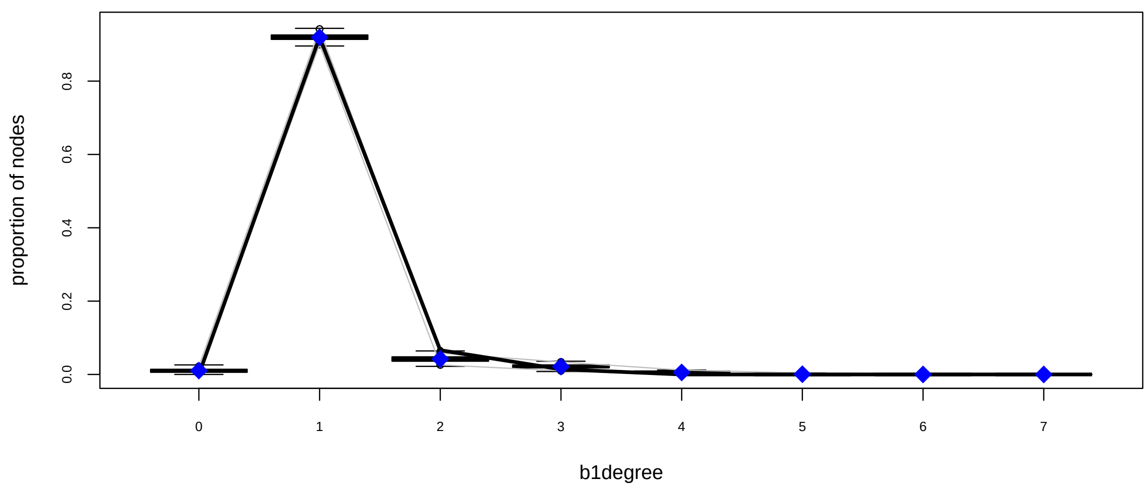
```
                        2.5% 25% 50% 75% 97.5%
edges                    -13  -4   1   6    16
b1degree1                -11  -4   0   4    10
cycle4                   -34 -13   0  16    52
b1factor.b1_gender.male  -10  -3   1   4    12


Are sample statistics significantly different from observed?
                  edges   b1degree1       cycle4 b1factor.b1_gender.male
diff.      1.253071e+00 -0.05457599 2.616475e+00            7.120689e-01
test stat. 1.837232e+01 -1.22226000 1.394755e+01            1.216653e+01
P-val.     2.188414e-75  0.22160932 3.256092e-44            4.685805e-34
                 (Omni)
diff.                NA
test stat.  4.981756e+02
P-val.     2.180719e-104

Sample statistics cross-correlations:
                             edges  b1degree1     cycle4
edges                    1.0000000 -0.6273622  0.8549354
b1degree1               -0.6273622  1.0000000 -0.5698004
cycle4                   0.8549354 -0.5698004  1.0000000
b1factor.b1_gender.male  0.5314239 -0.3247677  0.4217988
                        b1factor.b1_gender.male
edges                                 0.5314239
b1degree1                            -0.3247677
cycle4                                0.4217988
b1factor.b1_gender.male               1.0000000

Sample statistics auto-correlation:
Chain 1
              edges   b1degree1     cycle4 b1factor.b1_gender.male
Lag 0    1.00000000  1.00000000 1.00000000              1.00000000
Lag 640  0.46219220  0.36123841 0.36808454              0.56555488
Lag 1280 0.22083629  0.13173911 0.15681895              0.32971930
Lag 1920 0.10106233  0.05114607 0.06465253              0.19505666
Lag 2560 0.04911140  0.02504884 0.03203135              0.10243262
Lag 3200 0.01765148 -0.01676550 0.03047180              0.04651129
Chain 2
                edges  b1degree1     cycle4 b1factor.b1_gender.male
Lag 0     1.000000000 1.00000000 1.00000000              1.00000000
Lag 640   0.437858208 0.38674645 0.35549994              0.54439105
Lag 1280  0.170365470 0.17315370 0.11391348              0.30861060
Lag 1920  0.077068682 0.07864354 0.04139655              0.19067311
Lag 2560  0.014848922 0.03272962 0.01527084              0.11125867
Lag 3200 -0.002735924 0.02799260 0.01029926              0.06692818
Chain 3
                edges  b1degree1     cycle4 b1factor.b1_gender.male
Lag 0    1.0000000000 1.00000000 1.00000000              1.00000000
Lag 640  0.4638303468 0.38499285 0.36101793              0.56588893
Lag 1280 0.2486813189 0.18855567 0.18681347              0.33965285
Lag 1920 0.1317563283 0.10151423 0.09475260              0.19857586
Lag 2560 0.0475566842 0.05226031 0.04985226              0.09518536
Lag 3200 0.0003148158 0.02992456 0.02826998              0.01726816
Chain 4
                edges  b1degree1     cycle4 b1factor.b1_gender.male
Lag 0    1.000000e+00 1.00000000 1.00000000              1.00000000
Lag 640  4.474375e-01 0.40827558 0.36602612              0.56673910
Lag 1280 1.999899e-01 0.18376229 0.16108594              0.33121846
Lag 1920 7.810462e-02 0.06819393 0.06662650              0.17515468
Lag 2560 2.214036e-02 0.03229893 0.02759365              0.07527084
Lag 3200 1.327434e-05 0.03020447 0.01418929              0.04068117
Chain 5
                edges   b1degree1      cycle4 b1factor.b1_gender.male
Lag 0     1.000000000  1.00000000  1.00000000              1.00000000
Lag 640   0.437103052  0.38423058  0.34784445              0.56856874
Lag 1280  0.216170483  0.17224360  0.15877064              0.33285961
```

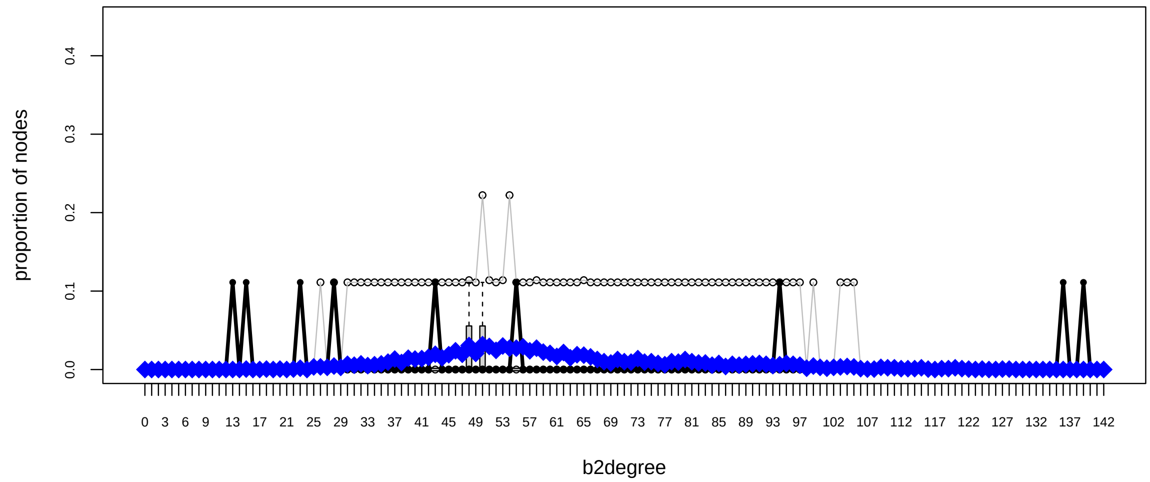
```
Lag 1920  0.105458281  0.07302384  0.09177188              0.19717275
Lag 2560  0.044444789  0.02504269  0.04868227              0.08931366
Lag 3200 -0.009955747 -0.01250088 -0.01686271              0.04036875
Chain 6
              edges    b1degree1     cycle4 b1factor.b1_gender.male
Lag 0    1.00000000  1.000000000 1.00000000              1.00000000
Lag 640  0.45435011  0.361268517 0.35812343              0.56960682
Lag 1280 0.21566779  0.153981676 0.15965314              0.33619428
Lag 1920 0.12654181  0.072802022 0.09339018              0.18847000
Lag 2560 0.05006227  0.007440694 0.03046906              0.09862146
Lag 3200 0.02497662 -0.008779587 0.01561748              0.02754053
Chain 7
               edges   b1degree1       cycle4 b1factor.b1_gender.male
Lag 0    1.000000000 1.000000000  1.000000000              1.00000000
Lag 640  0.463362102 0.362904627  0.363305483              0.56274889
Lag 1280 0.207687456 0.144460485  0.156659826              0.30735588
Lag 1920 0.089675932 0.044654202  0.078071052              0.17663718
Lag 2560 0.036751121 0.023897805  0.035773901              0.10232906
Lag 3200 0.003172149 0.006837233 -0.001948617              0.07518207
Chain 8
              edges  b1degree1      cycle4 b1factor.b1_gender.male
Lag 0    1.00000000 1.00000000 1.000000000              1.00000000
Lag 640  0.44705209 0.40603636 0.341088943              0.55333751
Lag 1280 0.22469509 0.18881850 0.160573033              0.31794405
Lag 1920 0.10221498 0.08973941 0.058634998              0.19089898
Lag 2560 0.03224681 0.04772215 0.027979386              0.10318507
Lag 3200 0.01803967 0.02104698 0.001062492              0.05033198
Chain 9
              edges   b1degree1       cycle4 b1factor.b1_gender.male
Lag 0    1.00000000 1.000000000  1.000000000              1.00000000
Lag 640  0.44119476 0.365874321  0.343237943              0.54715732
Lag 1280 0.21378712 0.165901421  0.149457193              0.29771860
Lag 1920 0.09606213 0.056225445  0.054445652              0.15065216
Lag 2560 0.03676788 0.018721924 -0.002869194              0.06468348
Lag 3200 0.03169008 0.002035173  0.007041516              0.02444377

Sample statistics burn-in diagnostic (Geweke):
Chain 1

Fraction in 1st window = 0.1
Fraction in 2nd window = 0.5

                  edges               b1degree1                  cycle4
             -0.8376964              -0.4855318              -0.5127717
b1factor.b1_gender.male
             -0.3354309

Individual P-values (lower = worse):
                  edges               b1degree1                  cycle4
              0.4022012               0.6272992               0.6081110
b1factor.b1_gender.male
              0.7373001
Joint P-value (lower = worse):  0.8182054
Chain 2

Fraction in 1st window = 0.1
Fraction in 2nd window = 0.5

                  edges               b1degree1                  cycle4
             -2.2283677              -0.1273937              -1.7492323
b1factor.b1_gender.male
             -1.1146404

Individual P-values (lower = worse):
                  edges               b1degree1                  cycle4
             0.02585601              0.89862881              0.08025087
b1factor.b1_gender.male
             0.26500453
```

```
Joint P-value (lower = worse):  0.09741188
Chain 3

Fraction in 1st window = 0.1
Fraction in 2nd window = 0.5

                edges             b1degree1                cycle4
           0.12534550           -0.54659601           -0.09595021
b1factor.b1_gender.male
          -0.46754728

Individual P-values (lower = worse):
                edges             b1degree1                cycle4
            0.9002500             0.5846563             0.9235601
b1factor.b1_gender.male
            0.6401084
Joint P-value (lower = worse):  0.8205227
Chain 4

Fraction in 1st window = 0.1
Fraction in 2nd window = 0.5

                edges             b1degree1                cycle4
           -0.8731893             0.4147736            -0.6828871
b1factor.b1_gender.male
           -1.0626114

Individual P-values (lower = worse):
                edges             b1degree1                cycle4
            0.3825599             0.6783077             0.4946782
b1factor.b1_gender.male
            0.2879582
Joint P-value (lower = worse):  0.8571621
Chain 5

Fraction in 1st window = 0.1
Fraction in 2nd window = 0.5

                edges             b1degree1                cycle4
           -0.7303596             0.4544881            -0.7449774
b1factor.b1_gender.male
           -0.7221944

Individual P-values (lower = worse):
                edges             b1degree1                cycle4
            0.4651704             0.6494776             0.4562854
b1factor.b1_gender.male
            0.4701749
Joint P-value (lower = worse):  0.9284842
Chain 6

Fraction in 1st window = 0.1
Fraction in 2nd window = 0.5

                edges             b1degree1                cycle4
           -0.4456012            -0.3455419            -0.7847546
b1factor.b1_gender.male
           -0.2160909

Individual P-values (lower = worse):
                edges             b1degree1                cycle4
            0.6558853             0.7296870             0.4325975
b1factor.b1_gender.male
            0.8289169
Joint P-value (lower = worse):  0.7642974
Chain 7

Fraction in 1st window = 0.1
```

```
Fraction in 2nd window = 0.5

                 edges              b1degree1                 cycle4
            -0.3819243              0.7027031             -0.7415056
b1factor.b1_gender.male
             1.4539718

Individual P-values (lower = worse):
                 edges              b1degree1                 cycle4
             0.7025175              0.4822408              0.4583870
b1factor.b1_gender.male
             0.1459541
Joint P-value (lower = worse):  0.5891684
Chain 8

Fraction in 1st window = 0.1
Fraction in 2nd window = 0.5

                 edges              b1degree1                 cycle4
          -0.104999617           -0.968118379           -0.004073132
b1factor.b1_gender.male
          -0.599161923

Individual P-values (lower = worse):
                 edges              b1degree1                 cycle4
             0.9163761              0.3329853              0.9967501
b1factor.b1_gender.male
             0.5490649
Joint P-value (lower = worse):  0.6925802
Chain 9

Fraction in 1st window = 0.1
Fraction in 2nd window = 0.5

                 edges              b1degree1                 cycle4
             1.2922259             -0.9945804              1.0602979
b1factor.b1_gender.male
             1.4268769

Individual P-values (lower = worse):
                 edges              b1degree1                 cycle4
             0.1962789              0.3199404              0.2890091
b1factor.b1_gender.male
             0.1536154
Joint P-value (lower = worse):  0.5098452

Note: To save space, only one in every 5 iterations of the MCMC sample
  used for estimation was stored for diagnostics. Sample size per chain
  was originally around 17865 with thinning interval 128.

Note: MCMC diagnostics shown here are from the last round of
  simulation, prior to computation of final parameter estimates.
  Because the final estimates are refinements of those used for this
  simulation run, these diagnostics may understate model performance.
  To directly assess the performance of the final model on in-model
  statistics, please use the GOF command: gof(ergmFitObject,
  GOF=~model).
```


proportion of statistics
model statistics
edges
b1degree1
cycle4
b1factor.b1_gender.male
0.0
0.2
0.4
0.6
0.8
1.0


proportion of nodes
b1degree
0
1
2
3
4
5
6
7
0.0
0.2
0.4
0.6
0.8


proportion of nodes
b2degree
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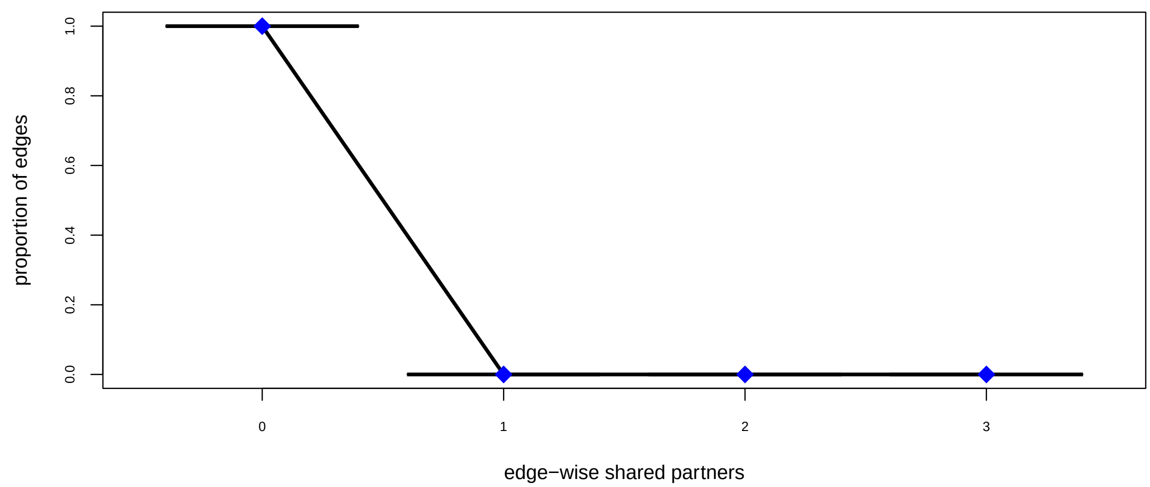
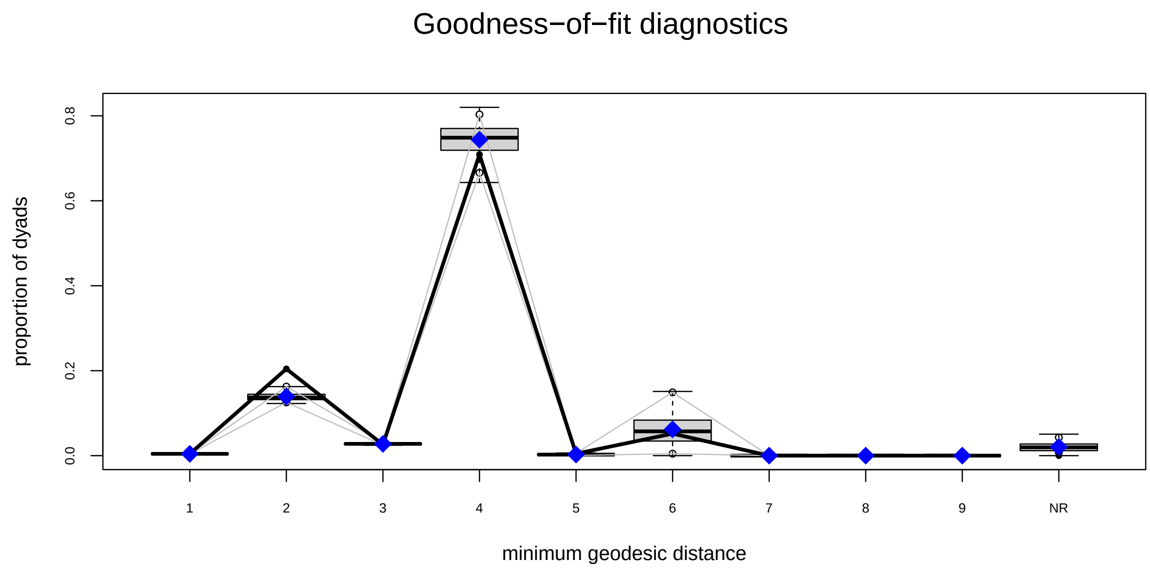
0.1
0.2
0.3
0.4


proportion of edges
edge−wise shared partners
0
1
2
3



Goodness-of-fit for model statistics

	obs	min	mean	max	MC p-value
edges	546	527	546.070	570	1.00
b1degree1	459	441	458.810	474	1.00
cycle4	67	20	67.250	144	0.99
b1factor.b1_gender.male	257	243	256.995	274	1.00

Goodness-of-fit for minimum geodesic distance

	obs	min	mean	max	MC p-value
1	546	527	546.070	570	1.00
2	26301	15809	17922.080	22664	0.00
3	3351	2921	3563.925	3912	0.26
4	91309	79282	95788.770	105587	0.34
5	594	0	333.755	953	0.18
6	6677	0	7976.075	24134	0.92
Inf	0	0	2647.325	6513	0.01

Goodness-of-fit for bipartition 1 degree

	obs	min	mean	max	MC p-value
0	0	0	5.250	13	0.01
1	459	441	458.810	474	1.00
2	33	9	21.095	32	0.00
3	7	3	10.715	18	0.30
4	0	0	2.755	7	0.13
5	0	0	0.345	2	1.00
6	0	0	0.030	1	1.00

Goodness-of-fit for bipartition 2 degree

	obs	min	mean	max	MC p-value
13	1	0	0.000	0	0.00
15	1	0	0.005	1	0.01
18	0	0	0.005	1	1.00
20	0	0	0.005	1	1.00
22	0	0	0.005	1	1.00
23	1	0	0.015	1	0.03
25	0	0	0.030	2	1.00
26	0	0	0.030	1	1.00
27	0	0	0.025	1	1.00

28	1	0	0.040	2	0.07
29	0	0	0.025	1	1.00
30	0	0	0.060	1	1.00
31	0	0	0.050	1	1.00
32	0	0	0.065	2	1.00
33	0	0	0.045	1	1.00
34	0	0	0.055	1	1.00
35	0	0	0.065	1	1.00
36	0	0	0.085	2	1.00
37	0	0	0.120	2	1.00
38	0	0	0.080	2	1.00
39	0	0	0.130	2	1.00
40	0	0	0.120	2	1.00
41	0	0	0.125	2	1.00
42	0	0	0.140	1	1.00
43	1	0	0.175	2	0.33
44	0	0	0.135	1	1.00
45	0	0	0.170	2	1.00
46	0	0	0.215	3	1.00
47	0	0	0.175	2	1.00
48	0	0	0.275	2	1.00
49	0	0	0.185	2	1.00
50	0	0	0.285	2	1.00
51	0	0	0.265	2	1.00
52	0	0	0.220	2	1.00
53	0	0	0.270	2	1.00
54	0	0	0.245	4	1.00
55	1	0	0.245	2	0.46
56	0	0	0.260	2	1.00
57	0	0	0.215	2	1.00
58	0	0	0.245	2	1.00
59	0	0	0.200	2	1.00
60	0	0	0.185	2	1.00
61	0	0	0.150	2	1.00
62	0	0	0.195	2	1.00
63	0	0	0.135	2	1.00
64	0	0	0.170	2	1.00
65	0	0	0.165	2	1.00
66	0	0	0.145	2	1.00
67	0	0	0.115	1	1.00
68	0	0	0.090	2	1.00
69	0	0	0.075	1	1.00
70	0	0	0.115	2	1.00
71	0	0	0.090	2	1.00
72	0	0	0.085	1	1.00
73	0	0	0.125	1	1.00
74	0	0	0.085	1	1.00
75	0	0	0.090	2	1.00
76	0	0	0.070	1	1.00
77	0	0	0.055	1	1.00
78	0	0	0.090	2	1.00
79	0	0	0.085	1	1.00
80	0	0	0.115	2	1.00
81	0	0	0.090	1	1.00
82	0	0	0.075	1	1.00
83	0	0	0.075	1	1.00
84	0	0	0.050	1	1.00
85	0	0	0.070	1	1.00
86	0	0	0.035	1	1.00
87	0	0	0.060	1	1.00
88	0	0	0.050	1	1.00
89	0	0	0.060	1	1.00
90	0	0	0.065	1	1.00
91	0	0	0.070	2	1.00
92	0	0	0.060	1	1.00
93	0	0	0.045	2	1.00
94	1	0	0.055	1	0.11
95	0	0	0.070	1	1.00

```
96    0    0 0.060   1        1.00
97    0    0 0.050   1        1.00
98    0    0 0.015   1        1.00
99    0    0 0.040   1        1.00
100   0    0 0.025   1        1.00
101   0    0 0.015   1        1.00
102   0    0 0.025   1        1.00
103   0    0 0.030   1        1.00
104   0    0 0.035   1        1.00
105   0    0 0.030   1        1.00
106   0    0 0.010   1        1.00
107   0    0 0.005   1        1.00
108   0    0 0.005   1        1.00
109   0    0 0.025   1        1.00
110   0    0 0.020   1        1.00
111   0    0 0.020   1        1.00
112   0    0 0.010   1        1.00
113   0    0 0.010   1        1.00
114   0    0 0.010   1        1.00
115   0    0 0.020   1        1.00
116   0    0 0.005   1        1.00
118   0    0 0.010   1        1.00
119   0    0 0.010   1        1.00
120   0    0 0.020   1        1.00
121   0    0 0.010   1        1.00
122   0    0 0.005   1        1.00
124   0    0 0.005   1        1.00
127   0    0 0.005   1        1.00
128   0    0 0.005   1        1.00
136   1    0 0.000   0        0.00
139   1    0 0.000   0        0.00

Goodness-of-fit for edgewise shared partner

       obs        min       mean        max MC p-value
    546.00     527.00     546.07     570.00       1.00
```

A.4.5 Model 4

This section contains only the console output for the GOF diagnostics for Model 4. The main diagnostics are contained in the Results section.

```
Goodness-of-fit for model statistics

                          obs   min      mean   max MC p-value
edges                     546   528   546.755   567       0.93
b1degree1                 459   445   459.310   472       0.95
cycle4                     67    29    68.370   148       0.98
b1factor.b1_gender.male   257   244   257.430   280       1.00
b1cov.b1_age            20959 20177 20992.900 21751       0.92

Goodness-of-fit for minimum geodesic distance

      obs   min      mean    max MC p-value
1     546   528   546.755    567       0.93
2   26301 15719 18052.580  22225       0.00
3    3351  2939  3586.890   3925       0.27
4   91309 78517 96179.115 105613       0.32
5     594     0   313.345    948       0.16
6    6677     0  7631.630  24797       0.96
Inf     0   507  2467.685   6513       0.00

Goodness-of-fit for bipartition 1 degree

  obs min    mean max MC p-value
0   0   1   4.890  13       0.00
1 459 445 459.310 472       0.95
2  33  10  20.650  33       0.01
3   7   3  10.900  18       0.22
```

4	0	0	2.840	9	0.11
5	0	0	0.380	3	1.00
6	0	0	0.025	1	1.00
7	0	0	0.005	1	1.00

Goodness-of-fit for bipartition 2 degree

	obs	min	mean	max	MC p-value
13	1	0	0.000	0	0.00
15	1	0	0.000	0	0.00
22	0	0	0.005	1	1.00
23	1	0	0.010	1	0.02
24	0	0	0.005	1	1.00
25	0	0	0.005	1	1.00
26	0	0	0.015	1	1.00
27	0	0	0.010	1	1.00
28	1	0	0.030	1	0.06
29	0	0	0.065	2	1.00
30	0	0	0.035	2	1.00
31	0	0	0.020	1	1.00
32	0	0	0.050	1	1.00
33	0	0	0.040	1	1.00
34	0	0	0.060	2	1.00
35	0	0	0.070	2	1.00
36	0	0	0.110	2	1.00
37	0	0	0.075	2	1.00
38	0	0	0.090	1	1.00
39	0	0	0.110	2	1.00
40	0	0	0.175	2	1.00
41	0	0	0.160	2	1.00
42	0	0	0.235	2	1.00
43	1	0	0.255	2	0.48
44	0	0	0.245	2	1.00
45	0	0	0.260	2	1.00
46	0	0	0.250	2	1.00
47	0	0	0.260	2	1.00
48	0	0	0.215	2	1.00
49	0	0	0.230	2	1.00
50	0	0	0.245	2	1.00
51	0	0	0.240	2	1.00
52	0	0	0.240	2	1.00
53	0	0	0.210	3	1.00
54	0	0	0.215	2	1.00
55	1	0	0.225	2	0.40
56	0	0	0.195	2	1.00
57	0	0	0.205	2	1.00
58	0	0	0.190	2	1.00
59	0	0	0.180	2	1.00
60	0	0	0.125	2	1.00
61	0	0	0.170	2	1.00
62	0	0	0.130	2	1.00
63	0	0	0.165	2	1.00
64	0	0	0.095	1	1.00
65	0	0	0.105	3	1.00
66	0	0	0.090	1	1.00
67	0	0	0.095	2	1.00
68	0	0	0.090	2	1.00
69	0	0	0.095	2	1.00
70	0	0	0.070	1	1.00
71	0	0	0.040	1	1.00
72	0	0	0.060	1	1.00
73	0	0	0.080	2	1.00
74	0	0	0.080	1	1.00
75	0	0	0.125	2	1.00
76	0	0	0.110	2	1.00
77	0	0	0.130	2	1.00
78	0	0	0.145	1	1.00
79	0	0	0.085	1	1.00

```
80    0    0 0.075   1       1.00
81    0    0 0.070   1       1.00
82    0    0 0.065   1       1.00
83    0    0 0.095   1       1.00
84    0    0 0.080   2       1.00
85    0    0 0.105   2       1.00
86    0    0 0.055   1       1.00
87    0    0 0.070   1       1.00
88    0    0 0.090   1       1.00
89    0    0 0.080   1       1.00
90    0    0 0.035   1       1.00
91    0    0 0.065   1       1.00
92    0    0 0.060   1       1.00
93    0    0 0.060   1       1.00
94    1    0 0.060   1       0.12
95    0    0 0.035   2       1.00
96    0    0 0.025   1       1.00
97    0    0 0.030   1       1.00
98    0    0 0.030   1       1.00
99    0    0 0.035   1       1.00
100   0    0 0.050   2       1.00
101   0    0 0.035   1       1.00
102   0    0 0.035   1       1.00
103   0    0 0.025   1       1.00
104   0    0 0.015   1       1.00
105   0    0 0.060   2       1.00
106   0    0 0.020   1       1.00
107   0    0 0.020   1       1.00
108   0    0 0.025   1       1.00
109   0    0 0.005   1       1.00
110   0    0 0.005   1       1.00
111   0    0 0.040   2       1.00
112   0    0 0.010   1       1.00
113   0    0 0.015   1       1.00
114   0    0 0.015   1       1.00
115   0    0 0.005   1       1.00
116   0    0 0.015   1       1.00
117   0    0 0.015   1       1.00
118   0    0 0.015   1       1.00
120   0    0 0.010   1       1.00
121   0    0 0.005   1       1.00
122   0    0 0.005   1       1.00
124   0    0 0.005   1       1.00
126   0    0 0.005   1       1.00
129   0    0 0.005   1       1.00
136   1    0 0.000   0       0.00
139   1    0 0.000   0       0.00

Goodness-of-fit for edgewise shared partner

      obs        min       mean        max MC p-value
  546.000    528.000    546.755    567.000      0.930
```

A.5 Source Code - Data Preprocessing

```r
# ---------------------------------------------------------------------------- #
# If not already installed
install.packages("viridis")  # For Colours
install.packages("here")     # To locate files from RProj

# Ensure repeatability
set.seed(42)
# ---------------------------------------------------------------------------- #
# Import the Bondora P2P Dataset
obj_paths = "resources/objects/"
bondora_raw <- read.csv("dataset/LoanData_Bondora.csv")
raw_cols <- colnames(bondora_raw)
# ---------------------------------------------------------------------------- #
# Select Columns to Keep
keep_cols <- c("LoanId", "UserName","Age", "Gender",
               "Country", "Amount", "Interest","LoanDuration",
              "UseOfLoan", "Rating", "Restructured", "MonthlyPayment",
              "OccupationArea", "BiddingStartedOn","Country")

bondora <- bondora_raw[keep_cols]

# Change date format to the correct one
bondora$BiddingStartedOn <- as.POSIXct(bondora$BiddingStartedOn,
                                       format = "%Y-%m-%d %H:%M:%S")
bondora$year <- as.numeric(format(bondora$BiddingStartedOn, "%Y"))

# Remove Rows with any NAs -> Complete Dataset Preferred
print(paste("NA Count |",sum(is.na(bondora)),"rows"))
bondora <- na.omit(bondora)
print(paste("NA Count |",sum(is.na(bondora)),"rows"))

# Observe the distribution of Loan Use over Time
unique_counts <- tapply(bondora$UserName,
                        list(bondora$year, bondora$UseOfLoan),
                        function(x) length(unique(x)))
unique_counts[is.na(unique_counts)] <- 0

barplot(t(unique_counts),           # transpose so bars = loan types
        beside = TRUE,              # side-by-side bars
        col = viridis::viridis(ncol(unique_counts)),
        legend.text = colnames(unique_counts),
        args.legend = list(x = "topright", cex = 0.8),
        xlab = "Year",
        ylab = "Number of Unique Users")
user_counts_plt <- recordPlot()
saveRDS(user_counts_plt,
        here::here("resources","objects","preprocessing","user_cnt_plt.Rds"))

# Restrict to Recent Time Period
bondora_test <- bondora[bondora$year > 2013 & bondora$year < 2017, ]

# See the distribution of LoanUses
see_counts <- function(var1, var2) {
  counts <- table(var1, var2)
  total_counts <- colSums(counts)
  prop <- total_counts / sum(total_counts)

  return(list(total_counts, round(prop,2)))
}
before_counts <- see_counts(bondora_test$UserName, bondora_test$UseOfLoan)
barplot(before_counts[[1]])

# Number of unique users in the subsample
unique_users <- unique(bondora_test$UserName)
sample_users <- sample(unique_users, 500)
```

```r
bondora_clean <- bondora_test[bondora_test$UserName %in% sample_users, ]

after_counts <- see_counts(bondora_clean$UserName, bondora_clean$UseOfLoan)
barplot(after_counts[[1]])

# DEPRECATED METHOD
# Filter People with 2+ Loans
#user_counts <- table(bondora_test$UserName)
#multi_users <- names(user_counts[user_counts > 2])
#bondora_clean <- bondora_test[bondora_test$UserName %in% multi_users, ]

# DEPRECATED METHOD
# Remove Users with only One Loan
#user_counts <- table(bondora_clean$UserName)
#multi_users <- names(user_counts[user_counts > 5])
#bondora_clean <- bondora_clean[bondora_clean$UserName %in% multi_users, ]
#bondora_test <- bondora_raw[bondora_raw$UserName %in% multi_users, ]

# See if Ratings are Properly Encoded
unique(bondora_clean$Rating)
bondora_clean <- bondora_clean[bondora_clean$Rating != "", ]

# Extract UseofLoan Types and Turn into Factor
bondora_clean$UseOfLoan_factor <- as.factor(bondora_clean$UseOfLoan)
unique(bondora_clean$UseOfLoan_factor)

loan_use_labels <- c(
  "-1" = "NA",
  "0" ="Loan Consolidation",
  "1" = "Real Estate",
  "2" = "Home Improvement",
  "3" = "Business",
  "4" = "Education",
  "5" = "Travel",
  "6" = "Vehicle",
  "7" = "Other",
  "8" = "Health",
  "110" = "Other Business",
  "102" = "Undefined Business",
  "108" = "Undefined Business"
)
# Change Labels for Cleaned Dataset
bondora_clean$UseOfLoan_factor <- loan_use_labels[
  as.character(bondora_clean$UseOfLoan)]

# Change Labels for Uncleaned Dataset
bondora_test$UseOfLoan_factor <- loan_use_labels[
  as.character(bondora_test$UseOfLoan)]

# Add labels to the OccupationArea Variable
levels(as.factor(bondora_clean$OccupationArea)) # view codes

occupation_labels <- c(
  "-1" = "NA",
  "1" = "Other",
  "2" = "Mining",
  "3" = "Processing",
  "4" = "Energy",
  "5" = "Utilities",
  "6" = "Construction",
  "7" = "Retail",
  "8" = "Transport",
  "9" = "Hospitality",
  "10" = "Info/Telcom",
  "11" = "Finance",
  "12" = "Realestate",
```

```r
  "13" = "Research",
  "14" = "Admin",
  "15" = "Civil/Mil",
  "16" = "Education",
  "17" = "Healthcare",
  "18" = "Social",
  "19" = "Art/Ent",
  "20" = "Agriculture",
  "21" = "Forestry/Fish"
)
# store original
bondora_clean$occupation_code <- bondora_clean$OccupationArea
bondora_clean$occupation_label <- occupation_labels[
  as.character(bondora_clean$OccupationArea)]

bondora_test$occupation_code <- bondora_test$OccupationArea
bondora_test$occupation_label <- occupation_labels[
  as.character(bondora_test$OccupationArea)]
# ---------------------------------------------------------------------------- #
# Observe Descriptive Statistics

cols <- viridis::viridis(30)

# Make function to save plots
save_plot <- function(plt_nam) {
  plt <- recordPlot()
  saveRDS(plt, here::here("resources","objects","preprocessing",
                     paste0(plt_nam,".Rds")))
}

# Make function to consistently plot comparisons
plot_desc_hists <- function(df1, df2, col_name, type) {

  par(mfrow=c(1,3))

  hist(df1[[col_name]], xlab=type, col=cols[15], main="", breaks=10)
  mtext("Initial Sample", side=3, adj=0, line=0.25, cex=1, font=2)

  hist(df2[[col_name]], xlab=type, col=cols[15], main="", breaks=10)
  mtext("Reduced Sample", side=3, adj=0, line=0.25, cex=1, font=2)

  qqplot(df1[[col_name]], df2[[col_name]], main="", cex=1,
         xlab="Full Sample", ylab="Subsample", line=0.25)
  abline(0, 1, lty=2)

  mtext("QQ Plot", side=3, adj=0, line=0.25, cex=1, font=2)

  mtext(type, outer = TRUE, line = -2, side=3, cex = 1.3, font = 2)

  # Reset plot window
  par(mfrow=c(1,1), mar=c(5,4,4,2)+0.1)
}

plot_desc_bar <- function(df1, df2, col_name, type) {

  par(mfrow=c(1,2), mar=c(5,10,4,2))

  barplot(sort(table(df1[[col_name]]), decreasing = F),
          xlab=type, col=cols[15], horiz=TRUE, las=1)
  mtext("Initial Sample", side=3, adj=0, line=0.25, cex=1, font=2)

  barplot(sort(table(df2[[col_name]]), decreasing = F),
          xlab=type, col=cols[15], horiz=TRUE, las=1)
  mtext("Reduced Sample", side=3, adj=0, line=0.25, cex=1, font=2)

  mtext(type, outer = TRUE, line = -2, side=3, cex = 1.3, font = 2)
```

```r

  # Reset plot window
  par(mfrow=c(1,1), mar=c(5,4,4,2)+0.1)
}

plot_desc_hists(bondora_test, bondora_clean, "Amount","Amount")
save_plot("hist_amt")

plot_desc_hists(bondora_test, bondora_clean, "Interest","Interest")
save_plot("hist_int")

plot_desc_hists(bondora_test, bondora_clean, "LoanDuration","Loan Duration")
save_plot("hist_loandur")

plot_desc_hists(bondora_test, bondora_clean, "MonthlyPayment","Monthly Payment")
save_plot("hist_monpmt")

plot_desc_hists(bondora_test, bondora_clean, "Age","Age")
save_plot("hist_age")

plot_desc_bar(bondora_test, bondora_clean,"UseOfLoan_factor","Loan Purpose")
save_plot("bar_loanuse")

plot_desc_bar(bondora_test, bondora_clean,"Rating","Credit Rating")
save_plot("bar_rating")

plot_desc_bar(bondora_test, bondora_clean,"occupation_label","Occupation")
save_plot("bar_occupation")

plot_desc_bar(bondora_test, bondora_clean, "Country","Country")
save_plot("bar_country")

# Get Tabular Summary Statistics
tab_comps <- function(df1, df2, cols) {
  stats <- c("Mean"=mean, "Median"=median, "Std. Dev."=sd, "Min"=min, "Max"=max)

  get_stats <- function(d) {
    t(sapply(d[cols], function(x)
      sapply(stats, function(f) round(f(x, na.rm=TRUE), 2))
    ))
  }

  df1_stats <- get_stats(df1)
  df2_stats <- get_stats(df2)

  out <- rbind(
    cbind(DataFrame = "Initial Sample",
          Variable = rownames(df1_stats), df1_stats),
    cbind(DataFrame = "Reduced Sample",
          Variable = rownames(df2_stats), df2_stats)
  )
  rownames(out) <- NULL
  as.data.frame(out)
}

tab_results <- tab_comps(bondora_test, bondora_clean,
                         c("Amount","Interest","Age","LoanDuration"))

knitr::kable(tab_results)

# Save table for use in the Report
saveRDS(tab_results,
        file=paste0(obj_paths,"preprocessing/","summary_table.Rds"))
# --------------------------------------------------------------------------- #
# Convert Dataset into Incidence Matrix to form Network Object (for ERGM)
bondora_slim <- bondora_clean
```

```r

# Create the Incidence Matrix for Use of Loan
bondora_matrix <- table(
  bondora_slim$UserName, bondora_slim$UseOfLoan)
bondora_matrix[bondora_matrix > 0] <- 1 # Given that ergm.counts fails with GOF

# Create network object with counts as Edge attribute
bondora_net <- network::network(
  bondora_matrix, directed=FALSE, bipartite=nrow(bondora_matrix),
  ignore.eval = FALSE, names.eval="frequency", loops=FALSE)

# Set the bipartite Attribute - UNNCESSARY GIVEN bipartite=length(n)
len <- dim(bondora_matrix)[1]
len_b2 <- dim(bondora_matrix)[2]
b_indicator <- c(rep(1,len),rep(2,len_b2))
network::set.vertex.attribute(bondora_net,
                              "bipartite",value=b_indicator)

# Extract Partition 2 Labels
loan_use <- levels(bondora_clean$UseOfLoan_factor)

# Create Loan Type Attribute for Partition 2
b2_loantype <- rep(NA, len)
b2_loantype <- c(b2_loantype, loan_use)

if (length(b2_loantype) == network::network.size(bondora_net)) {
  network::set.vertex.attribute(
    bondora_net, "b2_loantype", value = b2_loantype)
  }

# Add Age Vertex Attribute to B1
age <- bondora_clean$Age[match(
  rownames(bondora_matrix), bondora_clean$UserName)]
b1_age <- c(age, rep(NA, len_b2))
network::set.vertex.attribute(
  bondora_net, "b1_age", value=b1_age
)

# Add Gender Vertex Attribute to B1
gender <- bondora_clean$Gender[match(
  rownames(bondora_matrix), bondora_clean$UserName)]
unique(gender) # Check to see if encoded properly
gender <- ifelse(gender == 0, "male","female")
b1_gender <- c(gender, rep(NA, len_b2))
network::set.vertex.attribute(
  bondora_net, "b1_gender", value = b1_gender
)

# Save the network and data frame object
saveRDS(bondora_slim, file=paste0(obj_paths,"preprocessing/","bondora_df.Rds"))
saveRDS(bondora_net, file=paste0(obj_paths,"preprocessing/","bondora_net.Rds"))
# --------------------------------------------------------------------------- #
# First, create Incidence Matrix again but with Frequency Counts
loan_use_matrix <- table(
  bondora_slim$UserName, bondora_slim$UseOfLoan)

# Get the Adjacency Matrix for Loan Use Similarity (Dependent QAP Variable)
adj_mat_loan_use <- loan_use_matrix %*% t(loan_use_matrix)
# Remove self weights to remove any loops
diag(adj_mat_loan_use) <- 0

# Get the Adjacency Matrix for Credit Rating Similarity (Predictor in QAP)
incidence_rating <- table(bondora_slim$UserName, bondora_slim$Rating)
adj_mat_rating <- incidence_rating %*% t(incidence_rating)
diag(adj_mat_rating) <- 0

```

```r
# Get Adjacency Matrix for Occupation Similarity (Predictor in QAP, Binary)
incidence_occupation <- table(bondora_slim$UserName,
                              bondora_slim$occupation_label)
adj_mat_occupation <- incidence_occupation %*% t(incidence_occupation)
adj_mat_occupation[adj_mat_occupation > 0] <- 1
diag(adj_mat_occupation) <- 0

# Get the Adjacency Matrix for Loan Amount (Control in QAP) - BINARY
# First, bin the Loan Amounts
bondora_slim$Amount_bins <- cut(
  bondora_slim$Amount, breaks=c(0,2000,4000,6000,8000,10000),
  labels = c(1:5)
)
incidence_amount_bins <- table(bondora_slim$UserName, bondora_slim$Amount_bins)
adj_mat_amount_bins <- incidence_amount_bins %*% t(incidence_amount_bins)
diag(adj_mat_amount_bins) <- 0

# Continous Absolute Difference Approach
avg_amount <- tapply(bondora_slim$Amount, bondora_slim$UserName, mean)
adj_mat_amount_diff <- outer(avg_amount, avg_amount,
                             FUN = function(x,y) abs(x - y))
diag(adj_mat_amount_diff) <- 0

# Get Matrix for Differences in Age
incidence_age <- table(bondora_slim$UserName, bondora_slim$Age)
borrower_ages <- as.numeric(colnames(incidence_age)[max.col(incidence_age)])
names(borrower_ages) <- rownames(incidence_age)
adj_mat_age <- outer(borrower_ages, borrower_ages,
                     FUN = function(x, y) abs(x - y))
rownames(adj_mat_age) <- colnames(adj_mat_age) <- names(borrower_ages)

# Get Adjacency Matrix for (same) Gender
incidence_gender <- table(bondora_slim$UserName, bondora_slim$Gender)
adj_mat_gender <- incidence_gender %*% t(incidence_gender)
# Make the matrix binary for homophily
adj_mat_gender <- ifelse(adj_mat_gender > 0, 1, 0)
diag(adj_mat_gender) <- 0

# Get Adjacency Matrix for Differences in Average Loan Duration
borrower_loandur <- sapply(tapply(bondora_slim$LoanDuration,
                                  bondora_slim$UserName, unique),
                           mean)
adj_mat_loandur_diff <- outer(borrower_loandur, borrower_loandur,
                              FUN=function(x,y) abs(x-y))
diag(adj_mat_loandur_diff) <- 0

# Get Adjacency Matrix for Homophily in Restructure of Loans
incidence_restructure <- table(bondora_slim$UserName, bondora_slim$Restructured)
adj_mat_rest <- incidence_restructure %*% t(incidence_restructure)
diag(adj_mat_rest) <- 0

# Save the objects for the QAP Regression in different Script
qap_paths = paste0(obj_paths,"/qap/")
saveRDS(b_indicator, file=paste0(
  "resources/objects/preprocessing/indicator.Rds"))

saveRDS(adj_mat_loan_use, file=paste0(qap_paths,"adj_mat_loanuse.Rds"))
saveRDS(adj_mat_occupation, file=paste0(qap_paths,"adj_mat_occup.Rds"))
saveRDS(adj_mat_rating, file=paste0(qap_paths,"adj_mat_rating.Rds"))
saveRDS(adj_mat_amount_diff, file=paste0(qap_paths,"adj_mat_amtdiffs.Rds"))
saveRDS(adj_mat_age, file=paste0(qap_paths,"adj_mat_agediffs.Rds"))
saveRDS(adj_mat_gender, file=paste0(qap_paths,"adj_mat_gender.Rds"))
saveRDS(adj_mat_loandur_diff, file=paste0(qap_paths,"adj_mat_loandurdiffs.Rds"))
saveRDS(adj_mat_rest, file=paste0(qap_paths,"adj_mat_rest.Rds"))
# ---------------------------------------------------------------------------- #
```

scripts/bondora_preprocessing.R

A.6 Source Code - QAP Linear Regression

```r
# ---------------------------------------------------------------------------- #
# If not already Installed
install.packages("viridis")   # For Colours
install.packages("here")      # To locate files from RProj
install.packages("gridExtra") # For multiple ggplot2 plots

# Set colour palette
cols <- viridis::viridis(30)

# Ensure repeatability
set.seed(42)
# ---------------------------------------------------------------------------- #
# Load Relevant Files
qap_path="resources/objects/qap/"

loan_use_mat <- readRDS(paste0(qap_path,"adj_mat_loanuse.RDS"))
rating_mat <- readRDS(paste0(qap_path,"adj_mat_rating.RDS"))
amt_diffs_mat <- readRDS(paste0(qap_path,"adj_mat_amtdiffs.RDS"))
age_diffs_mat <- readRDS(paste0(qap_path,"adj_mat_agediffs.RDS"))
gender_mat <- readRDS(paste0(qap_path,"adj_mat_gender.RDS"))
loandur_diffs_mat <- readRDS(paste0(qap_path,"adj_mat_loandurdiffs.RDS"))
rest_mat <- readRDS(paste0(qap_path,"adj_mat_rest.RDS"))
occup_mat <- readRDS(paste0(qap_path,"adj_mat_occup.Rds"))
# ---------------------------------------------------------------------------- #
# Create function to determine significance from t-value statistic
t_to_stars <- function(t) {
  stars <- rep("", length(t))
  stars[abs(t) >= 1.96]  <- "*"
  stars[abs(t) >= 2.576] <- "**"
  stars[abs(t) >= 3.291] <- "***"

  return(stars)
}

# Make function to save plots
save_qap_plot <- function(plt_nam) {
  plt <- recordPlot()
  saveRDS(plt, here::here("resources","objects","qap",
                          paste0(plt_nam,".Rds")))
}

# Make function to run QAPs automatically
run_QAP <- function(y_mat, xlist, varnames, model_name) {

  # Run QAP Linear Regression
  model <- sna::netlm(y = y_mat, x = xlist, nullhyp = "qapspp", reps = 3000)
  model$names <- varnames
  summary(model)

  # Obtain results from Model and Name them
  results <- model$coefficients
  names(results) <- varnames

  # Add significance stars to model results
  results_sig <- paste(round(results,3), t_to_stars(model$tstat))

  # Save Model
  saveRDS(model, here::here("resources","objects","qap",
                            paste0(model_name,".Rds")))

  # Prepare function output
  list_names <- c("model","results","results_sig")
  items <- list(model, results, results_sig)
  names(items) <- list_names

```

```r
  return(items)
}

# Make function to Plot QAP Coefficients
plot_coef_qap <- function(model,
                          model_name = "Model Estimates",
                          sub = "Subtitle",
                          x_axis_label = "Coefficient Estimate") {

  # Extract Coefficients and SEs from ERGM
  coefs <- model$coefficients
  ses <- model$coefficients / model$tstat

  # Make DF
  df <- data.frame(
    term = model$names,
    estimate = coefs,
    se = ses
  )

  df$lower <- df$estimate - 1.96 * df$se
  df$upper <- df$estimate + 1.96 * df$se

  df$sig <- df$lower * df$upper > 0
  df$sig <- as.character(df$sig) # Convert to character to make it work

  order_index <- order(df$estimate)
  df$term <- factor(df$term, levels = df$term[order_index])

  ggplot2::ggplot(df, ggplot2::aes(x = estimate, y = term)) +
    ggplot2::geom_vline(xintercept = 0, linetype = "dashed", color = "grey40") +
    ggplot2::geom_errorbarh(ggplot2::aes(xmin = lower, xmax = upper),
                            height = 0.2) +
    ggplot2::geom_point(size = 3) +
    ggplot2::theme_bw() +
    ggplot2::theme(
      panel.grid.major.x = ggplot2::element_blank(),
      panel.grid.minor.x = ggplot2::element_blank()) +
    ggplot2::labs(
      x = x_axis_label,
      y = "Variable",
      title = model_name,
      subtitle = sub
      #color = "Significance"
    )
}
# Make Function to Plot Diagnostics
# DEPRECATED SINCE PROGRESS MEETING 3
qap_diags <- function(model, title, filename) {
  resid <- model$model$residuals
  fitted <- model$model$fitted.values

  # Set viewing window to two plots
  par(mfrow=c(1,2))

  # Residuals Plot
  hist(resid, xlab="Residuals",
       main=title)

  # Fitted vs Residuals
  plot(fitted, resid, xlab="Fitted Values", ylab="Residuals",
       main=title)

  # Capture and Save the Plot
  save_qap_plot(filename)

```

```r
  # Reset plot view
  par(mfrow=c(1,1))
}

var_names <- c("Intercept", "Rating", "Occupation","Loan Amount", "Age",
               "Gender", "Loan Duration", "Restructured")
main_pred_names <- c("Intercept", "Rating", "Occupation")
main_pred_vars <- list(rating_mat, occup_mat)
all_pred_vars <- list(rating_mat, occup_mat, amt_diffs_mat, age_diffs_mat,
                      gender_mat, loandur_diffs_mat, rest_mat)
# ---------------------------------------------------------------------------- #
# Basic QAP Linear Regression 1 - Unstandardised + No Controls
qap_m1 <- run_QAP(loan_use_mat, main_pred_vars, main_pred_names, "qap_m1")
summary(qap_m1$model)

# Plot the QAP Model Coefficients and Save it
qap_m1_coef_plot <- plot_coef_qap(qap_m1$model,
                                  model_name = "MRQAP Model 1",
                                  sub = "No Controls")
saveRDS(qap_m1_coef_plot,
        here::here("resources","objects","qap","qap_m1_coef_plot.Rds"))

# Run Diagnostics Plots - DEPRECATED SINCE PROGRESS MEETING 3 SUGGESTIONS
#qap_diags(qap_m1,"QAP LR - Unstandardised + No Controls","qap_m1_diags")

# ---------------------------------------------------------------------------- #
# Basic QAP Linear Regression 2 - Unstandardised + Controls
qap_m3 <- run_QAP(loan_use_mat, all_pred_vars, var_names, "qap_m3")

summary(qap_m3$model)

# Plot the QAP Model Coefficients and Save it
qap_m3_coef_plot <- plot_coef_qap(qap_m3$model,
                                  model_name = "MRQAP Model 2",
                                  sub = "With Controls")
saveRDS(qap_m3_coef_plot,
        here::here("resources","objects","qap","qap_m3_coef_plot.Rds"))

# Arrange the Plots together
qap_coef_plots <- gridExtra::grid.arrange(qap_m1_coef_plot, qap_m3_coef_plot,
                        ncol = 2)
# Save the Plots
saveRDS(qap_coef_plots,
        here::here("resources","objects","qap","qap_coef_plots.Rds"))

# Run Diagnostics Plots - DEPRECATED SINCE PROGRESS MEETING 3 SUGGESTIONS
#qap_diags(qap_m3,"QAP LR - Unstandardised + Controls","qap_m3_diags")

# ---------------------------------------------------------------------------- #
# Collect Results in Table

# Extract R2 from the netlm object from the Console output
# It was too difficult to calculate R2 from scratch and netlm does not
# provide this as an output directly from the netlm object
calc_r2 <- function(model) {

  out <- capture.output(model)
  r2_line <- grep("Multiple R-squared", out, value = TRUE)
  r2 <- as.numeric(
    sub(".*Multiple R-squared:\\s*([0-9\\.]+).*", "\\1", r2_line))

  return(r2)
}

# Make function to get details of netlm manually
extract.netlm <- function(model) {
```

```r
  tr <- texreg::createTexreg(
    coef.names = model$names,
    coef = model$coefficients,
    se = model$coefficients / model$tstat,
    pvalues = model$pgreqabs,
    gof.names = c("R-squared"),
    gof = (calc_r2(model)),
    gof.decimal = c(TRUE)
  )
  return(tr)
}

# View the Table of Results
titles <- c("M1 | No Controls", "M2 | Controls")
extracted_models <- lapply(list(qap_m1$model, qap_m3$model), extract.netlm)
texreg::screenreg(extracted_models, custom.model.names = titles, digits = 3)

# Save the table for use in the Report
saveRDS(list(extracted_models, titles),
        here::here("resources","objects","qap","qap_tables.Rds"))
# -------------------------------------------------------------------------- #
```

scripts/qap_network_analysis.R

A.7 Source Code - ERGM Network Analysis

```r
# ---------------------------------------------------------------------------- #
# If not already Installed
install.packages("viridis")      # For Colours
install.packages("Rglpk")        # Additional solver for ERGMs
install.packages("here")         # To locate files from RProj
install.packages("gridExtra")    # To display multiple ggplots side by side

# Import the Network and Other Object
ergm_path <- "resources/objects/ergm/"
bondora_net <- readRDS(here::here(
  "resources","objects","preprocessing","bondora_net.Rds"))
b_indicator <- readRDS(here::here(
  "resources","objects","preprocessing","indicator.Rds"))

# Set colour palette
cols <- viridis::viridis(30)

# Determine acceptable core count
n_cores <- parallel::detectCores() - 3 # Leave some out for other processes
print(paste("You have",n_cores,"usable cores"))

# Repeatability
set.seed(42)

# Save plots
save_ergm_plot <- function(plt_nam) {
  plt <- recordPlot()
  saveRDS(plt, here::here("resources","objects","ergm",
                          paste0(plt_nam,".Rds")))
}
# ---------------------------------------------------------------------------- #
# Copy network for plotting
bondora_plot <- bondora_net

# Get node type for plotting
type_indicator <- ifelse(b_indicator == 2, TRUE,FALSE)
shape <- ifelse(type_indicator,"square","circle")
network::set.vertex.attribute(bondora_plot, "shape", shape)

# Get Category Count for Vertex Size
counts <- sna::degree(bondora_plot)
counts_att <- ifelse(type_indicator, log(counts)*4, counts*2.5)
network::set.vertex.attribute(bondora_plot, "size", counts_att)

# Colours for the Node Types
plot_cols <- ifelse(type_indicator, cols[5], cols[30])
network::set.vertex.attribute(bondora_plot, "color", plot_cols)

# Legend Plotting
type_legend <- ifelse(type_indicator, "Borrowers", "Loan Type")
type_legend <- as.factor(type_legend)

# Plot the Network
plot(snafun::to_igraph(bondora_plot),
     #main = "Bipartite User-LoanUse",
     edge.arrow.size = 0.3,
     edge.color = rgb(0,0,0, alpha = 0.35),
     vertex.frame.color = "black",
     vertex.label = NA,
     vertex.frame.size = 3,
     edge.curved = FALSE,
     layout=igraph::layout.fruchterman.reingold)
legend("bottomleft",
       legend = levels(type_legend),
       inset = c(0.15, 0.01),
```

```r
        col = c(cols[30], cols[5]),
        pch = c(16, 15),
        title = "Node Partitions",
        title.font = 2,
        cex = 1,               # Increase the text size
        pt.cex = 2,              # Increase the point symbol size
        box.lwd = 1,             # Thin box border
        box.col = "black",       # Box color
        bty = "o"                # Use a box around legend
)
save_ergm_plot("network_plot")

# Summary Statistics and Save Them
density <- snafun::g_density(bondora_net)[1]
centralization <- snafun::g_centralize(bondora_net)[1]
vertices <- snafun::count_vertices(bondora_net)[1]
edges <- snafun::count_edges(bondora_net)[1]
dist <- snafun::g_mean_distance(bondora_net)[1]

net_names <- c("Vertex Count","Edge Count","Density","Centralization",
               "Mean Distance")
net_stats <- c(vertices, edges, density, centralization, dist)
net_stats <- sapply(net_stats, function(x) round(as.numeric(x),2))

net_summary <- data.frame(Statistic = net_names,
                          "Measure" = net_stats)
knitr::kable(net_summary)
saveRDS(net_summary, here::here("resources","objects","ergm","net_summary.Rds"))

# Plot Network Summary Statistics
snafun::plot_centralities(bondora_net)
save_ergm_plot("network_plots")
# ------------------------------------------------------------------------ #
# Make function to calculate probabilities from log odds
lodds_to_prob <- function(l_odd) {
  return(exp(l_odd) / (1 + exp(l_odd)))
}
# Make function to save ERGM object
save_ergm <- function(object, id) {
  saveRDS(object, file=here::here(
    "resources","objects","ergm",paste0(id,".Rds")))
}
# Make function to conduct ERGMs automatically
auto_ergm <- function(model, mcmc, name) {

  # Conducts the GOF Diagnostics and then saves the model,
  # mcmc diagnostics and gof object in a list.
  # This list can be imported as an .RDS object into the R environment

  # Diagnostics
  if (mcmc) {
    ergm::mcmc.diagnostics(model)
  }

  # The GOF must be adjusted otherwise it takes too long
  # We do not limit the GOF by changing its range of parameters
  gof <- ergm::gof(model,
                   control = ergm::control.gof.ergm(
                     nsim = 200,
                     MCMC.burnin = 5000,
                     MCMC.interval = 1000,
                     parallel = n_cores,
                     parallel.type = "PSOCK"
                   ))

  # Return List to view each item separately
```

```
  result <- list(model, gof)
  names(result) <- c("model","gof")
  save_ergm(result, paste0(name,"_panel"))

  return(result)
}
# Make Function to do Coefficient Plot
plot_coef <- function(model,
                      model_name = "Model Estimates",
                      sub = "Subtitle",
                      x_axis_label = "Coefficient Estimate") {

  # Extract Coefficients and SEs from ERGM
  coefs <- coef(model)
  ses <- sqrt(diag(vcov(model)))

  # Make DF
  df <- data.frame(
    term = names(coefs),
    estimate = coefs,
    se = ses
  )

  df$lower <- df$estimate - 1.96 * df$se
  df$upper <- df$estimate + 1.96 * df$se

  df$sig <- df$lower * df$upper > 0
  df$sig <- as.character(df$sig) # Convert to character to make it work

  order_index <- order(df$estimate)
  df$term <- factor(df$term, levels = df$term[order_index])

  ggplot2::ggplot(df, ggplot2::aes(x = estimate, y = term)) +
    ggplot2::geom_vline(xintercept = 0, linetype = "dashed", color = "grey40") +
    ggplot2::geom_errorbar(ggplot2::aes(xmin = lower, xmax = upper),
                           height = 0.2) +
    ggplot2::geom_point(size = 3) +
    ggplot2::theme_bw() +
    ggplot2::theme(
      panel.grid.major.x = ggplot2::element_blank(),
      panel.grid.minor.x = ggplot2::element_blank()) +
    ggplot2::labs(
      x = x_axis_label,
      y = "Variable",
      title = model_name,
      subtitle = sub
      #color = "Significance"
    )
}
# ---------------------------------------------------------------------------- #
# Find max degree
(max_deg <- max(summary(bondora_net ~ b2factor("b2_loantype"))))
# ---------------------------------------------------------------------------- #
# Base Model + GOF
formula_base_model <- bondora_net ~ edges
base_ergm <- ergm::ergm(formula_base_model)

base_ergm_panel <- auto_ergm(base_ergm, mcmc = FALSE, name = "ergm_base")

snafun::stat_plot_gof(base_ergm_panel$gof)
models = list(base_ergm)
texreg::screenreg(models)
# ---------------------------------------------------------------------------- #
# Iteration 1 + MCMC Diagnostics + GOF
model_1_params <- bondora_net ~ edges +
  # See if individuals tend to pick one loan type
```

```
  b1degree(1)

model_1 <- ergm::ergm(
  model_1_params,
  #constraints = ~ bd(minout = 0, maxout = 80),

  control = ergm::control.ergm(
    # Greater burn-in for cleaner result
    MCMC.burnin = 20000,
    # Greater sample size for greater stability
    MCMC.samplesize = 100000,
    seed = 42,
    MCMC.interval = 1000,
    # Only needed for convergence pvals to improve
    MCMLE.maxit = 45,
    # Smaller steps for stability
    MCMLE.steplength = 0.25,
    parallel = n_cores,
    parallel.type = "PSOCK"
  )
)

model_1_panel <- auto_ergm(model=model_1, mcmc=TRUE, name="ergm_m1")
model_1_panel$gof

snafun::stat_plot_gof(model_1_panel$gof)
save_ergm_plot("m1_gof")

texreg::screenreg(list(base_ergm, model_1))

# Plot Coefficients
m1_coef_plot <- plot_coef(model_1, model_name = "Model 1",
                          sub = "b1degree")
saveRDS(m1_coef_plot,
        here::here("resources","objects","ergm","m1_coef_plot.Rds"))
# ---------------------------------------------------------------------------- #
# Iteration 2 + MCMC Diagnostics + GOF
model_2_params <- bondora_net ~ edges + b1degree(1) +
  # See if there is clustering around pairs of loan types
  cycle(4)

model_2 <- ergm::ergm(
  model_2_params,
  #constraints = ~ bd(minout = 0, maxout = 80),

  control = ergm::control.ergm(
    # Greater burn-in for cleaner result
    MCMC.burnin = 20000,
    # Greater sample size for greater stability
    MCMC.samplesize = 100000,
    seed = 42,
    MCMC.interval = 1000,
    # Only needed for convergence pvals to improve
    MCMLE.maxit = 45,
    # Smaller steps for stability
    MCMLE.steplength = 0.25,
    parallel = n_cores,
    parallel.type = "PSOCK"
    )
  )

model_2_panel <- auto_ergm(model=model_2, mcmc=TRUE, name="ergm_m2")
snafun::stat_plot_gof(model_2_panel$gof)
model_2_panel$gof
models <- list(base_ergm, model_1, model_2)
texreg::screenreg(models)
```

```r

# Plot Coefficients
m2_coef_plot <- plot_coef(model_2_panel$model,
                          model_name = "Model 2",
                          sub = "b1degree + cycle(4)")
saveRDS(m2_coef_plot,
        here::here("resources","objects","ergm","m2_coef_plot.Rds"))
# ---------------------------------------------------------------------------- #
# Iteration 3 + MCMC Diagnostics + GOF
model_3_params <- bondora_net ~ edges + b1degree(1) + cycle(4) +

  # See if there is gender effect
  b1factor("b1_gender")

model_3 <- ergm::ergm(
  model_3_params,
  #constraints = ~ bd(minout = 0, maxout = 80),

  control = ergm::control.ergm(
    # Greater burn-in for cleaner result
    # Cycles mixed with the rest of the terms requires more burnin samples
    MCMC.burnin = 20000,
    # Greater sample size for greater stability
    MCMC.samplesize = 100000,
    seed = 42,
    # Reducing interval to reduce complexity between intervals
    MCMC.interval = 1000,
    # Only needed for convergence pvals to improve
    MCMLE.maxit = 25,
    # Smaller steps for stability
    MCMLE.steplength = 0.25,
    parallel = n_cores,
    parallel.type = "PSOCK"
  )
)

model_3_panel <- auto_ergm(model=model_3, mcmc=TRUE, name="ergm_m3")
snafun::stat_plot_gof(model_3_panel$gof)
model_3_panel$gof
models <- list(base_ergm, model_1, model_2, model_3)
texreg::screenreg(models)

# Plot Coefficients
m3_coef_plot <- plot_coef(model_3_panel$model,
          model_name = "Model 3",
          sub = "b1degree + cycle(4) + b1factor(gender)")
# Save Plot
saveRDS(m3_coef_plot,
        here::here("resources","objects","ergm","m3_coef_plot.Rds"))
# ---------------------------------------------------------------------------- #
# Iteration 4 + MCMC Diagnostics + GOF
model_4_params <- bondora_net ~ edges + b1degree(1) + cycle(4) +
  b1factor("b1_gender") +

  # See if higher ages make a difference
  b1cov("b1_age")

model_4 <- ergm::ergm(
  model_4_params,
  #constraints = ~ bd(minout = 0, maxout = 80),

  control = ergm::control.ergm(
    # Greater burn-in for cleaner result
    MCMC.burnin = 20000,
    # Greater sample size for greater stability
    MCMC.samplesize = 100000,
```

```r
    seed = 42,
    MCMC.interval = 1000,
    # Only needed for convergence pvals to improve
    MCMLE.maxit = 45,
    # Smaller steps for stability
    MCMLE.steplength = 0.25,
    parallel = n_cores,
    parallel.type = "PSOCK"
  )
)

model_4_panel <- auto_ergm(model=model_4, mcmc=TRUE, name="ergm_m4")
model_4_panel$gof
snafun::stat_plot_gof(model_4_panel$gof)
models <- list(base_ergm, model_1, model_2, model_3, model_4)
texreg::screenreg(models)

# Plot Coefficients
m4_ergm_coef_plot <- plot_coef(model_4_panel$model,
                               model_name = "Model 4",
                               sub = "Unrestricted Model")
# Save Plot
saveRDS(m4_ergm_coef_plot,
        here::here("resources","objects","ergm","m4_coef_plot.Rds"))
# ---------------------------------------------------------------------------- #
# Collect Plots Together
gridExtra::grid.arrange(m1_coef_plot, m2_coef_plot,
                        m3_coef_plot, m4_ergm_coef_plot,
                        nrow = 2, ncol = 2)

# Collect Models in Table 1: Raw results
headers <- c("Base ERGM","b1degree(1)","cycle(4)","Age","Gender")
texreg::knitreg(models, custom.model.names = headers,
                                digits = 3)

# Save the Table to use in Report
saveRDS(list(models, headers),
        here::here("resources","objects","ergm","ergm_table.Rds"))

# Padding
pad <- function(x, max) {
  return(c(unname(x), rep("-", max-length(x))))
}

# Collect Models in Table 2: Log Odds Results
max_ergms <- length(model_4$coefficients)
df_prob <- data.frame(
  Terms = names(model_4$coefficients),
  "Base" = pad(round(lodds_to_prob(base_ergm$coefficients),3),max_ergms),
  "b1degree(1)" = pad(round(lodds_to_prob(model_1$coefficients),3),max_ergms),
  "cycle(4)" = pad(round(lodds_to_prob(model_2$coefficients),3),max_ergms),
  "Age" = pad(round(lodds_to_prob(model_3$coefficients),3),max_ergms),
  "Gender" = pad(round(lodds_to_prob(model_4$coefficients),3),max_ergms)
)
knitr::kable(df_prob)

# Save Table to use in Report
saveRDS(df_prob,
        here::here("resources","objects","ergm","ergm_prob_tables.Rds"))
# ---------------------------------------------------------------------------- #
```

scripts/ergm_network_analysis.R

B Technology Statement

During the preparation of this work, we used ChatGPT in order to generate select parts of the `R` script utilised to process the dataset. Specifically, the tool was used to transform the processed dataset into a format that `igraph` would accept as a network object. Additionally, some `R` functions written by the group were improved with ChatGPT's suggestions. No AI tool was utilised to independently write parts of the report. The following parts of the assignment were affected/generated by AI tool usage: **INTRODUCTION;** The tool was utilised to evaluate the validity of certain theoretical concepts and refine them. **DATASET;** the data described within this section was processed partly by some code drafted by ChatGPT and edited by the group. After using this tool/service, **Group 7** evaluated the validity of the tool's outputs, including the sources that generative AI tools have used, and edited the content as needed. As a consequence, **Group 7** takes full responsibility for the content of their work.